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**Predictive Modeling for Real Estate Price
Using Machine Learning Methods**

Master's thesis

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Year of defense: 2025

Declaration of Authorship

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Prague, July 29, 2025

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Abstract

This thesis evaluates how textual descriptions from online real estate listings can improve the prediction of apartment prices in the Czech Republic. A dataset of more than 1.3 million advertisements from 2014–2024 is enriched with locational and other features. The text is transformed into numerical representations using TF-IDF, Word2Vec and BERT embeddings and combined with structured data in regularised regression and ensemble learning models. Adding textual information reduces mean absolute error by up to five percent for Lasso and Ridge regressions, with the largest gains in rural areas. Random Forest and XGBoost deliver the highest overall accuracy but benefit little from descriptions, suggesting that they already capture most predictive signals from structured variables. The results demonstrate that simple models can be substantially enhanced by harnessing property descriptions while advanced ensembles remain dominant for mass appraisal.

JEL Classification C45, C55, R31

Keywords real estate valuation, machine learning, text mining, Czech Republic

Title Predictive Modeling for Real Estate Price Using Machine Learning Methods

Abstrakt

Tato práce zkoumá, zda textové popisy z internetových inzerátů mohou zlepšit predikci cen bytů v České republice. Více než 1,3 milionu inzerátů z let 2014–2024 je obohaceno o prostorové a další charakteristiky. Texty jsou převedeny pomocí TF-IDF, Word2Vec a BERT embeddingů a spojeny s daty v regularizovaných regresích i ensemble modelech. Zařazení textových informací snižuje střední absolutní chybu u Lasso a Ridge až o pět procent, nejvýrazněji ve venkovských oblastech. Random Forest a XGBoost dosahují nejlepší celkovou přesnost, ale z popisů těží minimálně, protože strukturovaná data obsahují většinu predikční informace. Výsledky ukazují, že jednoduché modely lze znatelně zlepšit využitím popisů, zatímco pokročilé ensemble algoritmy zůstávají nejpřesnějším nástrojem pro hromadné oceňování.

Klasifikace JEL C45, C55, R31

Klíčová slova oceňování nemovitostí, strojové učení, zpracování textu, Česká republika

Název práce Prediktivní modelování cen nemovitostí pomocí metod strojového učení

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Acronyms

ALE	Accumulated Local Effects
ANNs	Artificial Neural Networks
AVMs	Automatic Valuation Models
BERT	Bidirectional Encoder Representations from Transformers
CBOW	continuous bag of words
ČSÚ	Český statistický úřad
ČÚZK	Český úřad zeměměřický a katastrální
GIS	Geographic Information System
HPMs	hedonic pricing models
IDF	Inverse Document Frequency
IML	Interpretable Machine Learning
LASSO	Least Absolute Shrinkage and Selection Operator
LightGBM	Light Gradient Boosting Machine
LRMSE	Log Root Mean Squared Error
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MARS	Multivariate Adaptive Regression Splines
ML	Machine Learning
mmMAPE	Max-Min Mean Absolute Percentage Error
MSE	Mean Squared Error
NLP	Natural Language Processing
PCA	Principal Component Analysis
R^2	R-Squared

RF Random Forest

RMSE Root Mean Squared Error

RNN Recurrent Neural Networks

RÚIAN Registr územní identifikace, adres a nemovitostí

SVM Support Vector Machine

TF Term Frequency

TF-IDF Term Frequency-Inverse Document Frequency

Word2Vec Word to Vector

XAI Explainable AI

XGBoost Extreme Gradient Boosting

Master's Thesis Proposal

Author	Bc. Vojtěch Kania Štykar, M.Sc.
Supervisor	PhDr. Lenka Šťastná, Ph.D.
Proposed topic	Predictive Modeling for Real Estate Price Using Machine Learning Methods

Motivation Accurate real estate valuation is crucial for various stakeholders, including buyers, sellers, investors, and policymakers. Traditional valuation methods, often grounded in basic regression analysis, may not capture the complexities of the real estate market. Hedonic-based methods face limitations due to key assumptions and estimation challenges, including identifying supply and demand, market disequilibrium, variable selection, functional form of the equation, and market segmentation (Fan et al., 2006; Schulz and Werwatz, 2004; Selim, 2009).

Machine learning (ML) techniques offer advanced capabilities to enhance predictive accuracy. For example, Park and Bae (2015) found that AdaBoost significantly outperformed other methods in terms of predictive accuracy. Recent studies have further enhanced the prediction efficacy by incorporating property descriptions. For instance, Baur et al. (2023) showed that including textual descriptions in real estate valuation could decrease predictive errors by up to 17%. Similarly, Zhang et al. (2023) highlighted that combining textual and non-textual features significantly improves prediction performance. Additionally, Guo et al. (2020) emphasised the superiority of ML models, such as random forest, in predicting housing prices using text mining techniques over conventional regression approaches.

This thesis will compare the performance of ML models against basic regression analysis in predicting real estate prices. Building upon these findings, this thesis will uniquely assess the differential impact of incorporating unstructured property description data in rural versus urban areas, a perspective not thoroughly explored in existing literature. It will also aim to offer robust predictive models for practical use by comparing the performance of ML models against basic regression analysis in predicting real estate prices in the Czech Republic. Importantly, this will be the first

study to apply these advanced ML methods to the Czech real estate market, thus contributing original insights and methodologies.

Hypotheses

1. **Hypothesis #1:** Machine learning models have higher predictive accuracy than basic regression analysis for real estate price prediction.
2. **Hypothesis #2:** Ensemble models outperform individual machine learning models in predicting housing prices in the Czech Republic.
3. **Hypothesis #3:** Incorporating unstructured data (such as property descriptions) along with structured data (e.g., apartment size, location, features like balcony, kitchen, elevator) will enhance the predictive accuracy of machine learning models.
4. **Hypothesis #4:** The predictive accuracy improvement from incorporating unstructured property descriptions will be more significant for properties in rural areas, where housing offer density is lower, compared to urban areas.

Methodology The dataset for this research comprises over 400,000 real estate advertisements for apartments for sale collected from multiple Czech real estate portals, covering the period from 2014 to 2024. These advertisements include listings from all municipalities in the Czech Republic, except those with no recorded listings. Key variables in the dataset include the date of advertisement creation, last found on, apartment size, location, and specific features such as balcony, kitchen, and elevator. The focus will initially be on comparing the performance of ML algorithms with basic regression analysis. These models will be trained and tested using traditional evaluation metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 to assess predictive accuracy.

Furthermore, the study will examine the impact of integrating both structured and unstructured data on predictive accuracy. Natural Language Processing (NLP) techniques will be applied to extract features from the textual descriptions of properties. The performance of models incorporating both types of data will be compared to those using only structured data, employing the same evaluation metrics.

The methodology will also include a focused analysis to determine whether the improvement in predictive accuracy from incorporating unstructured property description data is more significant for properties in rural areas compared to urban areas. This will involve splitting the dataset into rural and urban subsets based on predefined criteria such as population density. For each subset, the study will compare the performance of machine learning models using both structured data and unstructured data (property descriptions) with models using only structured data.

The performance of models in each subset (rural and urban) will be evaluated using the same traditional evaluation metrics. The differential impact of incorporating property descriptions on predictive accuracy between rural and urban areas will thus be assessed.

Expected Contribution The main contribution of this thesis is to identify whether the predictive accuracy improvement, gained from incorporating unstructured property description data, is more significant for properties in rural areas compared to urban areas. Unlike previous studies, such as Zhang et al. (2024) which focused on improvement from property description in five selected cities in Canada and Guo et al. (2020), which emphasised urban contexts, this study will uniquely determine the differential impact of description data in rural versus urban real estate markets.

Additionally, this thesis will highlight the benefits of combining textual descriptions with structured data for more accurate predictions, as evidenced by recent research (Zhang et al., 2023; Guo et al., 2020). Furthermore, this research will demonstrate the overall effectiveness of ensemble ML models to simple ML models and effectiveness of machine learning algorithms in comparison to basic regression analysis with regards to predictive accuracy.

Outline

1. Introduction

- Background and significance of accurate real estate valuation
- Overview of machine learning methods
- Objectives and scope of the research

2. Literature Review

- Review of traditional real estate valuation methods
- Examination of previous studies on ML applications in real estate

3. Methodology

- Description of dataset
- Detailed methodology for testing each hypothesis

4. Data Analysis and Results

- Results from ML models versus traditional methods, including basic regression, comparison
- Performance of the ensemble models versus individual ML models

- Impact of unstructured data on prediction accuracy

5. Discussion

- Interpretation of key findings
- Implications for real estate market stakeholders
- Limitations and future research directions

6. Conclusion

- Summary of findings
- Contributions to academic literature and practical applications

Core bibliography

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Chapter 1

Introduction

Classical hedonic pricing models have been used for decades to explain housing prices since the seminal work of Rosen (1974). Meta-analyses highlight the importance of structural and locational attributes (Sirmans et al., 2006), yet linear specifications often struggle to capture market complexities, motivating further methodological developments (Khoshnoud, Sirmans, and Zietz, 2023). Comparative studies show that machine learning techniques can offer superior predictive accuracy (Kok, Koponen, and Martínez-Barbosa, 2017; Mayer et al., 2019). Recent research also suggests that incorporating information from listing descriptions provides additional gains (Baur, Rosenfelder, and Lutz, 2023; Bittencourt et al., 2022).

Real estate plays a pivotal role in modern economies and accurate price estimation is of interest to households, investors and policy makers alike. Traditional hedonic pricing models capture the impact of structural and locational attributes on property values, yet recent advances in data availability and computing power have spurred interest in applying machine learning techniques. Alongside structured features, online listings contain rich textual descriptions that may offer additional predictive power but are rarely exploited in large-scale studies.

The central question examined is whether the textual narratives contained in property advertisements improve valuation accuracy beyond conventional variables and whether this improvement differs between urban and rural markets. Previous research has largely concentrated on major cities and has rarely compared performance across the urban-rural divide (Baur, Rosenfelder, and Lutz, 2023; A.D. Nowak, Price, and P.S. Smith, 2021). Leveraging a nationwide dataset of over 1.3 million listings, including more than 200,000 observations

from municipalities with fewer than 10,000 inhabitants, this thesis is able to analyse these differences systematically.

This thesis investigates how information extracted from property descriptions can improve the prediction of residential prices in the Czech Republic. Using a dataset of over 1.3 million apartment listings from 2014–2024, it evaluates the contribution of textual features to both regularised regression and ensemble learning models. Building on the literature reviewed in Chapter 2, the analysis tests whether natural language processing techniques help reduce prediction errors, in particular for smaller rural markets where structured data are often limited.

The remainder of this thesis is organised as follows. Chapter 2 surveys existing approaches to real estate valuation and the integration of text data. Chapter 3 describes the modeling framework and outlines the algorithms considered. The construction of the dataset and the preprocessing steps are detailed in Chapter 4. Empirical results are presented in Chapter 5 and their implications are discussed in Chapter 6. Finally, Chapter 7 summarises the main findings and suggests avenues for further research.

Chapter 2

Literature Review

The purpose of this chapter is to review the existing body of research pertinent to real estate valuation, focusing on the evolution of methodologies employed to predict property prices. This review encompasses traditional approaches as well as contemporary advancements that leverage Machine Learning (ML) techniques and unstructured textual data. By systematically analysing the academic literature, this chapter provides a foundation for understanding the contributions and limitations of various approaches, contextualising the scope and objectives of this thesis.

This chapter is structured as follows. The first section examines the foundational role of hedonic pricing models (HPMs) in real estate valuation. It highlights the theoretical underpinnings, applications, and critiques of these models, while also discussing their evolution to address methodological limitations. The second section explores the growing body of research employing machine learning models in the context of real estate valuation. Particular attention is paid to the comparative performance of ML models relative to traditional regression techniques and the methodological innovations introduced by these models. The third and final section reviews studies that incorporate textual data into property valuation. This includes both regression-based approaches and advanced ML techniques that leverage Natural Language Processing (NLP) to extract valuable features from unstructured data such as property descriptions.

2.1 Hedonic Pricing Models

Hedonic pricing models (HPMs) have long been a cornerstone in the valuation of real estate, offering a systematic framework to decompose property prices into their constituent characteristics. Already in 1974, Rosen proposed in his seminar paper the hedonic model to explain the concept of hedonic prices and implicit markets of a differentiated commodity. According to Rosen (1974), the quality of a particular feature of a differentiated commodity influences the decision of the consumer. This framework conceptualises the price of a property as the aggregate value of its attributes.

Since Rosen (1974) the HPMs were used for real estate valuation in multiple studies (for a review, see Khoshnoud, Sirmans, and Zietz 2023). Sirmans et al. (2006) provided a meta-analysis of key housing attributes, confirming that structural variables such as square footage and the number of bedrooms consistently influence property prices across diverse markets. Additionally, the importance of location-specific factors, including neighbourhood quality and accessibility to services, has been emphasised in numerous studies.

Despite their widespread use, hedonic models are not without limitations. Issues such as the specification of functional forms, omitted variable bias, and difficulties in capturing market segmentation have drawn criticism (Malpezzi, 2002; Sheppard, 1999). Researchers like Khoshnoud, Sirmans, and Zietz (2023) highlight the evolution of HPMs to address these concerns, incorporating advanced econometric techniques such as spatial autoregressive models, quantile regressions, and difference-in-difference approaches. These enhancements aim to improve model robustness by accounting for spatial dependencies and heterogeneities in housing markets. Additionally, recent innovations have sought to augment traditional hedonic models with new data sources and computational methods. For instance, the integration of GIS data has enabled more granular analyses of locational attributes.

2.2 Machine Learning (ML) Models Used for Real Estate Pricing

Beyond traditional econometric approaches, recent research has increasingly explored the application of machine learning (ML) techniques for residential property valuation. While ML models share the core principle of decom-

posing property value into its constituent attributes-akin to HPMs-they differ significantly in methodology. Classic HPMs are typically parametric and rely on strong assumptions regarding linearity and functional form. By contrast, ML approaches are predominantly non-parametric, allowing them to flexibly capture complex, non-linear interactions among variables without predefined model structures (Khoshnoud, Sirmans, and Zietz, 2023). As such, ML models represent a methodological advancement rather than a direct extension of traditional HPMs.

Among the range of machine learning techniques applied to residential price prediction are Support Vector Machine (SVM), Multivariate Adaptive Regression Splines (MARS), and the Least Absolute Shrinkage and Selection Operator (LASSO) algorithm. In addition to standard Artificial Neural Networks (ANNs), researchers have explored more advanced architectures such as Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) models, particularly when handling time-dependent housing data. Some methodologies further enhance performance by integrating the outputs of base models-such as linear regression and Random Forest-into a meta-model, often a neural network, to generate final predictions.

Ensemble methods, which combine multiple base models to improve performance, are particularly prevalent. These include tree-based algorithms such as Random Forest (RF), Gradient Boosting (GB), Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), and Extremely Randomized Trees. Random Forest is described as a non-parametric ensemble method using uncorrelated decision trees, which helps reduce variance and overfitting. It has been noted for its potential suitability for mass appraisal. XGBoost, a form of Gradient Boosting, is frequently chosen and has shown promising results in estimating real estate values. These tree-based ensemble methods are considered effective at handling complex, potentially nonlinear relationships and are robust to variable scaling.

While parametric HPMs are valuable for understanding the drivers of price (inference), their ability for predicting the price of unseen properties (predictive tasks) is often limited. In contrast, ML techniques are less constrained by predefined model structures and offer greater flexibility. They can learn from the underlying data and iteratively refine their predictive capabilities, leading to remarkably accurate results in comparative studies.

Comparative studies consistently show that ML techniques yield better predictions than standard linear models (Chun Lin and Mohan, 2011; Kok, Kopo-

nen, and Martínez-Barbosa, 2017; Mayer et al., 2019). Non-linear ML models, such as Gradient Boosting, Support Vector Machine (SVM), and Random Forest (RF), clearly perform better than linear models like Elastic Net and linear regression. For example, Hong, Choi, and Kim (2020) found that a RF predictor achieved an average percentage deviation of only around 5.5% between predicted and actual market prices, whereas an OLS-based predictor resulted in almost 20% deviation. The probability of a predicted price being within 5% of the actual price was 72% for the RF model, compared to only about 17.5% for the OLS model.

This enhanced predictive power is largely attributed to the capacity of ML models to handle complex, potentially non-linear relationships and interactions between property features, which traditional linear models struggle to capture. Traditional HPMs often impose strong, potentially unrealistic assumptions like linearity and the separability of attribute effects (Zurada, Levitan, and Guan, 2011). This rigid structure can lead to reduced accuracy when applied to complex, real-world markets (Hong, Choi, and Kim, 2020). ML models, particularly tree-based methods like RF and XGBoost, are well-equipped to model non-linear links and the varying influence of variables across different market segments or contexts. They can explore complex patterns and interactions without the need for the modeler to explicitly pre-specify them.

Studies using Explainable AI (XAI) methods have revealed non-linear relationships for many features (such as age of construction, living area, and location) that are difficult to represent using traditional parametric or semi-parametric models (Krämer et al., 2023). These analyses can even challenge long-established rules of thumb by revealing varying patterns across different locations or property types. ML models can capture bundles of hedonic characteristics and how they interact to influence price, sometimes showing that their combined effect is greater than the sum of individual contributions (Lorenz et al., 2023). The data-driven nature of many ML approaches means they learn complex structures and relationships directly from the data, without requiring the modeler to explicitly describe every aspect of the market's complexity (Tarasov and Dessoulavy-Śliwiński, 2025). This contrasts with traditional models that require careful, manual specification by experts based on economic theory, which can be time-consuming and struggle with non-linearity.

Furthermore, ML models are seen as potentially more efficient for large-scale or real-time valuations. Automatic Valuation Models (AVMs), which commonly utilise ML techniques, can be trained on vast datasets and provide property

value estimates within seconds (Steurer, Hill, and Pfeifer, 2021). This makes them suitable for applications like mass appraisal, where many properties need to be valued efficiently.

Finally, a significant advantage, highlighted by recent research, is the ability of ML models to effectively integrate diverse data types, including unstructured data like textual property descriptions, alongside traditional structured numerical features.

In summary, ML models offer notable advantages in residential property valuation due to their superior predictive accuracy, achieved through their flexibility in handling complex non-linear relationships and interactions, their ability to learn directly from data, their efficiency in large-scale applications, and their capacity to integrate valuable information from diverse data sources, including unstructured text.

However, despite their predictive power, a significant disadvantage of many complex machine learning models is their lack of transparency and interpretability, often referred to as the "black box" problem. While traditional econometric models like OLS regression are comprehensible to humans, understanding the internal calculations and logic of complex ML applications can be difficult or impossible. This opacity means that while ML models can accurately predict prices, they often fail to provide a transparent account of how these predictions were derived. This "black box" character poses challenges for justifying decisions, gaining trust, and generating new insights in real estate markets.

To address this limitation, the field of Explainable AI (XAI) or Interpretable Machine Learning (IML) has emerged. XAI tools and frameworks, many of which are model-agnostic (applicable to various ML methods), aim to reveal the operational modes of these algorithms and make their actions more transparent (Holzinger et al., 2022). Techniques like visualizing description embeddings, using Shapley additive explanations, or Accumulated Local Effects (ALE) plots are being explored to interpret ML models in real estate contexts (Baur, Rosenfelder, and Lutz, 2023; Lorenz et al., 2023; Rico-Juan and Taltavull de La Paz, 2021). These methods help to mitigate the trade-off between predictive performance and explainability, increasing acceptance and supporting research into market drivers and non-linearities.

Besides explainability, several other challenges and disadvantages are associated with using machine learning for residential pricing:

- **Data Availability and Quality:** ML models require large volumes of rele-

vant and high-quality data for training to produce unbiased and dependable results. The performance heavily depends on the availability of sufficient and accurately labeled data. However, obtaining annotated data can be scarce, expensive, and time-consuming. Furthermore, approaches using detailed data like textual descriptions are primarily applicable to real estate offers from online platforms, as digital text may not be available for all properties (Baur, Rosenfelder, and Lutz, 2023). Data quality issues, such as missing features or data entry errors, can introduce bias and diminish the model’s informative value (Lorenz et al., 2023).

- **Computational Costs:** Training complex ML models, especially deep learning architectures or sophisticated ensemble methods, can involve significant computational requirements (Lorenz et al., 2023). Tasks like hyperparameter tuning, often done through methods like grid searches combined with cross-validation, add to this cost. Processing textual data with advanced models like Bidirectional Encoder Representations from Transformers (BERT) or even simpler Term Frequency-Inverse Document Frequency (TF-IDF) with large vocabularies can also demand considerable processing time (Bittencourt et al., 2022).
- **Model Complexity and Selection:** The real estate domain can be complex, involving structured data, potentially unstructured data (like text), and spatial dependencies. There is a wide and expanding list of ML algorithms to choose from (Steurer, Hill, and Pfeifer, 2021). Selecting the optimal algorithm, data representation methods (e.g., for text), and hyperparameters for a specific task is a challenge. The choice of appropriate performance metrics for model selection via cross-validation is also crucial and can significantly influence which model is deemed best (Steurer, Hill, and Pfeifer, 2021).

Overall, the literature reflects a significant trend towards adopting a wide array of machine learning models for residential property valuation, moving beyond traditional econometric approaches to leverage data-driven techniques for improved predictive performance, while also acknowledging the importance of model interpretability.

2.3 Integration of Textual Data into Real Estate Pricing Models

Research is exploring the use of machine learning with different data types, including the integration of structured numerical data with unstructured textual descriptions of properties. Mostly, property descriptions were used to enhance predictive accuracy. While traditional models primarily rely on structured attributes (e.g., size, number of rooms, age, location coordinates), textual descriptions contain rich, often unique information that is not captured in structured features.

Property descriptions, commonly found in real estate listings, are unstructured textual narratives written by agents or sellers to highlight the attributes, amenities, and unique selling points of a property. Beyond conveying factual details, these descriptions often employ persuasive and emotive language intended to influence buyer perceptions. They may implicitly communicate information about the neighbourhood, quality of construction, recent renovations, or lifestyle benefits that are not explicitly captured in structured data fields. As such, property descriptions represent a rich yet underutilised source of qualitative information that can complement traditional variables in real estate valuation. In recent years, an emerging body of research has explored how natural language processing techniques can extract meaningful features from these texts to enhance predictive modeling of property prices. This section reviews key studies that integrate textual data into real estate pricing models.

The integration of textual information into real estate pricing models has evolved, starting with basic keyword-based approaches and progressing to more advanced text mining and machine learning techniques. Early work began exploring the potential of analysing textual data in real estate classifieds. These studies recognised that the language used in descriptions might relate to the property's price. A fundamental step in processing text data is transforming it into a numerical representation that can be used by predictive models. This often involves tokenisation, which splits the text into individual words or phrases. Basic preprocessing steps may include converting text to lowercase, removing punctuation, and eliminating common stop words.

Some initial approaches involved manually selecting keywords from property descriptions and creating variables based on their presence. However, manually chosen keywords can be time-consuming and prone to human error.

Data-driven approaches were proposed to build a real estate dictionary and identify relevant tokens (words or phrases) without relying on a predefined set of manually chosen keywords. These tokens can proxy for actual features of the property, seller, or neighbourhood, and indicator variables for these tokens can be included in models. Studies using these methods found that text information contained "hidden characteristics" and could be used to improve pricing models. For example, a study by Abdallah and Khashan (2017) identified sample words and bi-grams that affected price positively or negatively, such as "investment" and "balcony" having a positive effect, while "bus" and "unfurnished" had a negative effect. A. Nowak and P. Smith (2017) used a tokenisation approach and incorporated indicator variables for tokens into a hedonic model, showing that listing descriptions provided information that improved performance.

A more advanced method for representing textual data numerically is Term Frequency-Inverse Document Frequency (TF-IDF). TF-IDF assigns a numerical value to each token (n-gram) based on how frequently it appears in a specific document (property description) compared to how common it is across the entire collection of documents. This weighting scheme helps highlight terms that are specific to a particular listing but not necessarily common across all listings. Several studies have successfully employed TF-IDF to convert real estate descriptions into numerical features for price prediction models.

Abdallah and Khashan (2017) utilised TF-IDF as part of their data mining process to exploit textual data in classifieds and found it helped improve prediction accuracy. Stevens (2014) also explored TF-IDF as a text mining technique for predicting selling price, asking price, and price fluctuation, comparing it with Bag-of-Words models. Cardenas et al. (2022) compared TF-IDF and Bag-of-Words representations for house price classification and found that TF-IDF representations of house descriptions achieved the best predictive performance. Their experiments showed that a logistic regression model processing TF-IDF data achieved higher test accuracy compared to one using only tabular data. Bittencourt et al. (2022) ran extensive experiments with different extraction methods and learning algorithms and concluded that binary TF-IDF was the most promising method for extracting relevant knowledge from textual information, outperforming more modern algorithms based on deep neural networks in their study. Bushuyev et al. (2024) also used TF-IDF for processing textual data to enhance the accuracy of rental price prediction, demonstrating that adding textual features significantly reduced the MSE when combined with numerical features.

More recent advancements in Natural Language Processing (NLP) have introduced sophisticated word embedding techniques that capture not just the frequency of words but also their semantic meanings and contextual nuances. Bidirectional Encoder Representations from Transformers (BERT) is highlighted as a current state-of-the-art method for representing text. Contextualized embeddings like BERT can assign geometrically closer vectors to words with similar meanings. They also capture the polysemic nature of words, meaning they can represent different meanings depending on the context. Several studies have begun applying BERT embeddings to real estate valuation problems. Baur, Rosenfelder, and Lutz (2023) evaluated various machine learning models using different text representations, including BERT embeddings. Their empirical evaluation, based on large datasets of rental apartments in Berlin and house purchase offers in Los Angeles, showed that a gradient boosting model with BERT embeddings achieved the lowest predictive errors. They reported that including the property description into the best model reduced the MAE by up to 17.09 percent. Zhang, Li, and Branco (2024) also explored the use of BERT, alongside TF-IDF and Word to Vector (Word2Vec), to process house description text for price prediction. While they found that combining textual and non-textual features generally improved performance, their research indicated that Word2Vec embedding was the best textual strategy, followed by TF-IDF and then BERT. They hypothesized that this unexpected result for BERT might be due to it not being trained on a real estate-specific corpus, suggesting that training a word embedding model on an industry-specific corpus could be vital. Bushuyev et al. (2024) also mentioned BERT as one of the advanced word vectorization methods used in recent trends for integrating textual data with numerical features in real estate pricing forecasts.

Overall, the body of research demonstrates a clear trend towards leveraging textual descriptions in real estate valuation models. These studies consistently show that including information extracted from property descriptions, beyond just structured features, significantly improves the accuracy of price prediction models. The methods for incorporating text have evolved from basic keyword extraction and frequency counts to more sophisticated techniques like TF-IDF and advanced embeddings such as BERT, each offering different levels of insight and performance depending on the specific application and dataset.

Table 2.1: Overview of selected studies using textual data for real estate price prediction.

Study	Real Estate Type(s)	Model(s)	Textual Data Techniques
Zhang, Li, and Branco (2024)	Asking prices of houses	SVR, RF, GB, DNN	TF-IDF, Word2Vec, BERT
Bushuyev et al. (2024)	Rental apartments	LightGBM	TF-IDF
Baur, Rosenfelder, and Lutz (2023)	Rental and house purchase offers	GB, SVM, Elastic Net, Linear Regression	Bag-of-Words, Dictionary, Doc2Vec, BERT
Cardenas et al. (2022)	House purchases	Logistic Regression, MLP	TF-IDF, Bag-of-Words, Zero-shot LLM
Bittencourt et al. (2022)	House price valuations	Extremely Randomized Trees, GB, RF, XGBoost, LightGBM	TF-IDF, Bag-of-Words, FastText, BERT
A.D. Nowak, Price, and P.S. Smith (2021)	House purchases	LASSO	Bag-of-Words
Liu, A.D. Nowak, and P.S. Smith (2020)	House purchases	LASSO	Sufficient dictionary
A. Nowak and P. Smith (2017)	House purchases	LASSO	Bag-of-Words
Abdallah and Khashan (2017)	House purchases	Linear Regression	TF-IDF
Stevens (2014)	Selling prices, asking prices, price fluctuations	Naive Bayes, SVM, GB, SGD	TF-IDF, Bag-of-Words

The existing studies that implement textual information (review in Table 2.1) primarily focuses on datasets from urban or metropolitan areas. Research has utilised data from specific cities such as Berlin and Los Angeles (Baur, Rosenfelder, and Lutz, 2023), Houston (Bushuyev et al., 2024), Porto Alegre (Bittencourt et al., 2022), Atlanta and Phoenix (Liu, A.D. Nowak, and P.S. Smith, 2020), and a selection of the most populous Canadian cities (Zhang, Li, and Branco, 2024). Even when studies discussed geographic variation, this was typically examined at the level of different cities, Metropolitan Statistical Areas (MSAs), or zip codes within these more populated regions (A. Nowak and P. Smith, 2017; A.D. Nowak, Price, and P.S. Smith, 2021). While these studies acknowledge that real estate markets are highly localized and that textual dictionaries may vary spatially, their empirical scope has remained largely confined to urban or densely populated areas.

2.4 Residential Real Estate Market in the Czech Republic

The Czech Republic offers a uniquely instructive setting for studying residential property markets. In little more than three decades the country has moved from a centrally planned housing system to one of the most vibrant—and volatile—markets in Central Europe. Rapid mortgage-market liberalisation after 2000, coupled with constrained new construction, produced some of the fastest post-transition price gains in the region. Successive external shocks—the 2008–09 Global Financial Crisis, the 2012–13 domestic recession, and, most recently, the COVID-19 pandemic—amplified these cycles and exposed persistent structural rigidities such as lengthy permitting procedures and short-run supply inelasticity. Together, these forces have generated pronounced affordability pressures, especially in Prague and Brno, and have prompted an expanding body of empirical work on price drivers, regional disparities, and potential overvaluation.

Because of this distinctive trajectory, Czech studies serve two complementary purposes in the international literature. First, they act as a natural laboratory for testing whether price determinants derived in mature Western European or North-American contexts travel to a post-socialist economy with different institutional legacies. Second, they illustrate how methodological choices—ranging from panel cointegration techniques and vector-error-

correction models to machine-learning-based mass appraisal—can be adapted to data environments characterised by rapid structural change and marked urban-rural heterogeneity.

The remainder of this section synthesises the main findings from these Czech-focused contributions, organising them around three recurrent themes: (i) macro-financial linkages between credit, income and prices; (ii) regional heterogeneity in drivers and elasticities; and (iii) recent methodological innovations that, for example, exploit web-scraped offer data.

Several key studies have investigated house prices and their determinants in the Czech Republic. Brůha, Hlaváček, and Komárek (2017) investigated whether property price movements have detectable effects on Czech households' consumption and saving decisions, focusing on three episodes of price movements (2002/2003, 2007/2008, and 2009/2010). They utilised Household Budget Statistics (HBS) combined with regional property transaction data from the Czech Statistical Office (CZSO). Their methodologies included regression analysis (OLS and robust regression) and the Propensity Score Matching (PSM) approach, employing elastic net regression for variable selection. Their findings suggested that property owners, on average, have a lower propensity to consume and a higher saving rate, which runs counter to the collateral channel theory.

Hejlová, Hlaváček, and Komárek (2017) developed a comprehensive method for assessing house price sustainability in the Czech Republic, advocating for the simultaneous use of multiple models. They proposed four distinct approaches: a general supply and demand model; a financial accelerator model linking house prices to GDP and household credit; an assessment of the economic sense of home ownership through analysis of the price-to-rent ratio and comparison of annual costs of renting versus owning; and a price-to-income ratio. These approaches utilised CZSO apartment transaction data and Institute for Regional Information (IRI) asking prices and rents. They employed the Vector Error Correction Model (VECM) for econometric modeling and the Hodrick-Prescott (HP) filter for statistical indicators to determine deviations from equilibrium. Their work culminated in an aggregate assessment method that non-arbitrarily weighted the estimates based on their correlation. The results indicated that Czech residential property prices were close to their equilibrium level or slightly overvalued in mid-2014.

Čermáková, Hromada, and Machová (2022) focused on identifying regional differences in the relationship between flat prices and net disposable income of

households across the 14 Czech regions from 2000 to 2017. They used publicly available data and applied a linear mixed model with a random intercept for the region and an AR(1) process for error terms. Their findings indicated that flat prices in Prague grew faster with increasing household disposable income compared to most other regions, making Prague prices highly sensitive.

Kalabiška and Hlaváček (2022) examined housing price behaviour and identified their determinants across Czech regions from 2000 to 2017. They analysed a wide range of supply, demand, demographic, and macroeconomic variables from CZSO and CNB data. Their methodology included a panel dynamic OLS (DOLS) estimator to describe long-term equilibrium and an error correction model (ECM) to capture short-run dynamics. They also performed extensive panel unit root tests and Granger causality tests. The study found that apartment prices were primarily driven by wages, unemployment, and building plot prices, with regional differences in the impact of these factors. For instance, building plot prices had an unexpected negative effect in low-income regions.

More recent studies by Hromada and Čermáková (2021) and Čermáková, Hromada, and Piecha (2022) have explored issues of housing unavailability and the impact of the COVID-19 pandemic on the Czech real estate market. They analyse selected pricing and technical parameters and housing affordability dynamics, often highlighting regional disparities. Their research frequently draws on primary statistical data mined by their proprietary EVAL software, which collects, analyses, and evaluates real estate advertising and cadastral data from across the Czech Republic. These studies use basic statistical methods and comparative analysis of offer prices and other market indicators. They note that recent price growth has been accelerated by central bank policy, inflation fears, and a lack of new supply, leading to a widening price-to-income gap and decreasing housing affordability, particularly in major cities like Prague and Brno.

Collectively, these studies on the Czech residential real estate market highlight a trend of increasing prices, particularly in major cities, influenced by a combination of demand-side factors (income, credit, demographics, speculative investment), supply constraints (slow permitting, limited new construction), and macroeconomic policies (central bank interest rates, pension reform issues). Methodologically, there's a strong reliance on panel data analysis, including OLS, DOLS, ECM, and GLS regressions, to capture regional and temporal dynamics. The use of unit root and cointegration tests (e.g., Levin-Lin-Chu, Im-Pesaran-Shin, Hadri, Kao) is common for establishing long-term

relationships and checking for stationarity, crucial for detecting price bubbles. Furthermore, Granger causality tests are employed to understand predictive relationships between variables like prices and rents. The development of proprietary data mining software (e.g., EVAL) and web parsing techniques is also a notable methodological innovation, providing unique, high-frequency data for analysis. These diverse methodological approaches underscore the complexity of analysing the Czech Republic's residential real estate market, aiming to provide comprehensive insights for both academics and policymakers.

This thesis, utilising data from the entire Czech Republic, including both urban and rural areas, presents a unique and significant extension to this body of literature. By collecting and analysing data across the full urban-rural spectrum of a country, this work addresses a notable gap in the geographic coverage of previous studies implementing textual data. Real estate markets in rural areas often differ substantially from their urban counterparts in terms of property types, relevant amenities, neighbourhood descriptions, and overall market dynamics. For instance, while urban descriptions might emphasize proximity to public transport, or specific districts, rural descriptions might highlight features like land size, tranquility, or agricultural potential. Application of multiple textual data methods (TF-IDF, Word2Vec, and BERT) to this geographically diverse dataset, therefore, extends this literature. This allows for an investigation into how the effectiveness of these different text representation techniques varies across distinct market types within a single national context. It can reveal whether techniques like BERT, potentially sensitive to the specific language and context of the training corpus, perform differently when applied to descriptions from rural areas compared to urban ones, or how TF-IDF weights might shift to highlight features more relevant in one setting versus another.

Chapter 3

Methodology

This chapter outlines the methodological framework employed in this thesis, which aims to predict real estate prices using both traditional regression models and machine learning techniques. The analysis integrates structured property attributes and unstructured textual data extracted from property descriptions. By comparing Ridge and Lasso regression with ensemble machine learning methods such as Random Forest and XGBoost, this study investigates the extent to which textual information improves predictive accuracy, with a particular focus on differences between urban and rural housing markets.

3.1 Regression Models

Variable selection is traditionally an important step in developing regression-based models. However, the primary focus of this thesis is to examine the impact of natural language processing (NLP) methods on property valuation and their differential effects in urban versus rural areas. Therefore, manual feature selection is not a priority. Instead, this study employs Ridge and Lasso regression, which manage feature importance by regularisation. In other words, they automate the feature selection process.

By combining Ridge and Lasso regression, this study leverages the strengths of both techniques. Ridge regression ensures that all features are considered, but it addresses multicollinearity. Lasso regression simplifies the model by selecting the most influential predictors, enhancing interpretability.

3.1.1 Ridge Regression

Ridge regression is a linear regression technique that incorporates L2 regularisation. This regularisation adds a penalty term to the sum of squared residuals. This penalty is proportional to the squared magnitude of the coefficients, ensuring that the regression model does not assign excessively high weights to any predictors, even in the presence of multicollinearity. The Ridge regression objective function is expressed as:

$$\text{Objective Function} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \beta_j^2,$$

where:

- y_i and $\hat{y}_i$ represent the observed and predicted values, respectively,
- β_j are the regression coefficients,
- λ is the regularisation parameter that controls the degree of shrinkage applied to the coefficients.

Ridge regression does not eliminate features but reduces the magnitude of their coefficients. This reduction solves the multicollinearity problem while ensuring that all predictors contribute to the model.

3.1.2 Lasso Regression

Lasso regression is another regularised linear regression technique, but it applies L1 regularisation, which penalises the absolute values of the coefficients. The Lasso regression objective function is given by:

$$\text{Objective Function} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j|,$$

where:

- y_i and $\hat{y}_i$ represent the observed and predicted values, respectively,
- β_j are the regression coefficients,
- λ is the regularisation parameter that determines the strength of the penalty.

Unlike Ridge regression, Lasso regression can shrink some coefficients to zero. In other words, it can perform feature selection. This makes Lasso particularly suitable for identifying the most relevant predictors in datasets with many features.

3.2 Machine Learning Methods

Traditional regression models are used because they are simple and easily interpretable. They provide a clear understanding of the relationship between predictors and the target variable. However, they are limited in handling non-linear relationships and interactions. In contrast, ML methods excel in identifying complex patterns in data without requiring predefined relationships.

Machine learning (ML) methods have emerged as powerful tools for predictive modeling. They offer significant advantages over traditional regression-based techniques. While regression models rely on predefined functional relationships between features and the target variable, ML models by their design identify complex, nonlinear patterns in data. This flexibility makes them well-suited for datasets with high dimensionality and interactions between variables.

Although various machine learning methods exist, I chose to use XGBoost and Random Forest as representative machine learning models to compare with linear regression models. This decision is based on two primary reasons: first, the predictive accuracy and computational efficiency of XGBoost and Random Forest have been well established in prior research across different predictive tasks (e.g., Baur, Rosenfelder, and Lutz 2023 and Zhang, Li, and Branco 2024); second, the primary objective of this thesis is not to identify the best-performing machine learning algorithm, but rather to examine whether, and to what extent, the inclusion of textual data improves predictive performance in urban and rural contexts.

3.2.1 Random Forest (RF)

Random Forest (RF) is an ensemble learning method that combines multiple decision trees. It draws *bootstrap* samples of the training data for each tree (bagging), and randomly selects subsets of features at each split when building the tree.

Training Phase Let $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ be the training set, where $\mathbf{x}_i \in \mathbb{R}^d$ represents the feature vector of the i -th observation with d predictors (i.e., structured and textual features), and $y_i \in \mathbb{R}$ denotes the corresponding target value (log of price per m^2). A RF builds T decision trees independently, each tree trained on a bootstrap sample of size N drawn (with replacement) from the original dataset. Within each tree, at every split, a random subset of m features (out of d) is considered to find the best split. This mechanism reduces correlation between the trees and helps the ensemble generalise better.

Prediction Phase In regression, the final prediction for a new sample $\mathbf{x}^*$ is typically the average of the predictions from all T decision trees:

$$\hat{y}(\mathbf{x}^*) = \frac{1}{T} \sum_{t=1}^T h_t(\mathbf{x}^*),$$

where $h_t(\cdot)$ is the prediction function of the t -th decision tree. By averaging multiple trees - each with high variance but relatively low bias - Random Forests achieve strong overall performance and reduced variance compared to a single decision tree.

Regularization and Hyperparameters

- *Number of Trees* (**n_estimators**): Increasing the number of trees T generally improves performance and reduces variance, but comes with higher computational cost.
- *Maximum Depth* (**max_depth**): Limits how many splits each tree can have. Deeper trees are more expressive but can overfit.
- *Maximum Features* (**max_features**): Controls the number of candidate features considered at each split.
- *Bootstrap* (**bootstrap**): Determines whether sampling with replacement is used for building each tree. Setting this to **True** enables the standard bagging procedure.

3.2.2 Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) is a gradient boosting framework. It iteratively builds decision trees by minimising a loss function. Each subsequent tree focuses on correcting the errors of the previous ones.

Training Phase Given a training set $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ with $\mathbf{x}_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$ for regression, XGBoost sequentially adds trees to minimize a differentiable loss function $\mathcal{L}$. Let $\hat{y}_i^{(t-1)}$ be the prediction of the model (an ensemble of $t - 1$ trees) for the i -th sample. The idea is to add a new tree $h_t(\mathbf{x})$ that best fits the current residuals. Formally, the model after t iterations is:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta h_t(\mathbf{x}_i),$$

where η (the learning rate) scales the contribution of each new tree. XGBoost uses second-order gradient information to efficiently choose the structure of the new tree $h_t(\cdot)$, aiming to reduce the loss

$$\mathcal{L}^{(t)} = \sum_{i=1}^N \ell(y_i, \hat{y}_i^{(t-1)} + \eta h_t(\mathbf{x}_i)) + \Omega(h_t),$$

where ℓ is the loss (e.g., mean squared error for regression) and Ω is a regularisation term that penalises the complexity of the tree h_t . By iteratively optimising this objective, XGBoost balances reducing the error on training data with preventing overly complex trees.

Prediction Phase Once trained, the prediction for a new input $\mathbf{x}^*$ is the sum of the initial model (often a constant) and the contributions of all T trees:

$$\hat{y}(\mathbf{x}^*) = \sum_{t=1}^T \eta h_t(\mathbf{x}^*).$$

In this thesis, the final output is a real-valued prediction of *log of price per m^2* .

Regularization and Hyperparameters XGBoost provides extensive regularisation and hyperparameters that help control overfitting and improve performance:

- **Learning Rate (η):** Shrinks the contribution of each tree. Smaller values require more iterations but often yield better generalisation.
- **Number of Estimators (`n_estimators` or `num_boost_round`):** The total number of boosting stages (i.e., trees). More trees typically improve the model but increase training time and risk overfitting.

- **Maximum Depth** (`max_depth`): Controls the depth of each tree. Deeper trees can capture more complex patterns but they may overfit.
- **Regularization Terms** (`lambda`, `alpha`): λ (L2 regularisation) and α (L1 regularisation) penalise large weights in the tree leaves, helping reduce overfitting.
- **Gamma** (γ): A threshold for further splitting a leaf node. Larger values make the algorithm more conservative (fewer splits).
- **Subsample** (`subsample`): The fraction of the training data sampled (without replacement) before growing trees. Useful to reduce overfitting by adding randomness.
- **Colsample_bytree/Colsample_bylevel**: The fraction of features to be randomly sampled for each tree (and optionally each split). Reduces correlation between trees.

3.2.3 Grid Search

Hyperparameter optimisation is a critical component of developing high-performance ML models. Hyperparameters, unlike model parameters learned during training, are pre-defined and significantly influence the model's accuracy and generalisability. Grid search is a systematic method for hyperparameter tuning that evaluates all possible combinations of a predefined set of hyperparameter values to identify the best configuration.

The process entails:

1. **Definition of the Hyperparameter Grid:** Specification of a range of possible values for each hyperparameter.
2. **Training and Validation:** Evaluation of each hyperparameter combination using cross-validation to ensure results are generalisable across data subsets.
3. **Performance Measurement:** Usage of evaluation metrics to assess model performance.
4. **Selection of Optimal Parameters:** Choice of the hyperparameter combination that achieves the best performance on the validation set.

Hyperparameter optimisation was in most cases performed in two stages. In the first stage, a wide range of hyperparameter values was explored to identify promising regions of the hyperparameter space. The second stage focused on fine-tuning the hyperparameters around the best-performing configurations from the first stage, allowing for more precise optimisation. Hyperparameters used in this thesis are summarised in Table C.1 in Appendix. Cross-validation was employed to ensure robust model evaluation and to mitigate the risk of overfitting. Standard 5-fold cross-validation with shuffling was applied. Therefore, the model was trained on four folds and tested on the remaining fold, repeating the process five times. Averaging the results across folds provides a realistic measure of generalisability that is not overly influenced by a specific train-test split.

The key hyperparameters of the final trained models are summarised in Table C.2 in Appendix.

3.3 Evaluation Metrics

To assess the performance of the regression and ML models developed in this thesis, a set of evaluation metrics was employed. These metrics were selected based on Steurer, Hill, and Pfeifer (2021) who collected a set of 48 metrics and chose metrics well suitable for evaluating Automatic Valuation Models (AVMs). The following metrics were employed:

- **Mean Absolute Error (MAE):** MAE measures the average absolute difference between the predicted and actual values. It is calculated as:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|,$$

where y_i represents the actual value, $\hat{y}_i$ represents the predicted value, and n is the total number of observations. MAE is robust to outliers which makes it a reliable metric for assessing the overall predictive accuracy of the models.

- **Max-Min Mean Absolute Percentage Error (mmMAPE):** mmMAPE measures the percentage error between predicted and actual values, treat-

ing overpredictions and underpredictions symmetrically. It is defined as:

$$\text{mmMAPE} = \frac{1}{n} \sum_{i=1}^n \left(\frac{\max(y_i, \hat{y}_i)}{\min(y_i, \hat{y}_i)} - 1 \right).$$

Unlike Mean Absolute Error (MAE), which focuses on absolute errors, mmMAPE captures relative errors as a percentage. This is particularly useful in real estate valuation, where property prices can vary widely (e.g., urban vs rural properties). mmMAPE ensures predictions are evaluated proportionally to their magnitude.

mmMAPE addresses the known asymmetry issues of traditional Mean Absolute Percentage Error (MAPE), treating overpredictions and underpredictions equally. It is robust in cases where housing prices vary greatly, ensuring fair comparisons across different price ranges.

This thesis focuses on real estate valuation, where understanding proportional prediction accuracy (e.g., percentage error relative to price) is a valuable complement to absolute measures like MAE and variance-based measures like R-Squared (R^2). By incorporating mmMAPE, the evaluation framework is enriched with a metric that provides insights into the relative accuracy of model predictions across diverse property types and locations.

- **Log Root Mean Squared Error (LRMSE):** LRMSE measures the average squared difference between the logarithms of predicted and actual values, making it symmetric in dispersion and robust when evaluating models with log-transformed data. It is defined as:

$$\text{LRMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\log(y_i) - \log(\hat{y}_i))^2}.$$

LRMSE is particularly useful for predictions, where log transformations help normalise the target variable and stabilise variance.

- **R-Squared (R^2):** R^2 evaluates the proportion of variance in the dependent variable that is explained by the independent variables. It is given by:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2},$$

where $\bar{y}$ is the mean of the actual values. An R^2 value closer to 1 indicates a better fit, whereas a value close to 0 suggests that the model explains little of the variability in the data.

R^2 is included in this thesis to complement the Mean Absolute Error (MAE). While MAE provides a clear measure of error in absolute terms, introducing Root Mean Squared Error (RMSE) would result in redundancy, as both focus on error magnitudes. R^2 , on the other hand, adds a distinct layer of understanding by assessing how well the independent variables collectively explain the variability in housing prices. This makes it particularly valuable in the context of real estate valuation, where understanding the contribution of features to price prediction is as important as measuring prediction accuracy.

3.4 Dependent Variable

The choice of *log of price per m^2* as the dependent variable in this study is motivated by its ability to standardise property values across diverse property types and sizes. *Price per m^2* is a widely used benchmark in real estate research. Additionally, this metric is used in practice by investors and developers in the Czech Republic. Therefore, findings of this study could be also beneficial outside of academia.

3.5 Natural Language Processing of Property Descriptions

Property descriptions in real estate listings constitute a form of unstructured textual data that complements structured attributes such as size, location, and condition. These descriptions are typically written by sellers or agents to highlight key features, amenities, and qualities of the property, often using persuasive or emotive language to influence potential buyers. In addition to explicitly stated facts, descriptions may implicitly convey information about the neighbourhood, lifestyle, or perceived value of the property. By extracting and quantifying linguistic patterns from these texts, it is possible to incorporate qualitative information into predictive models.

3.5.1 TF-IDF

TF-IDF is a statistical measure that evaluates the importance of a word in a document relative to a corpus of documents. In this study, property description is treated as a document. TF-IDF was implemented since it is a widely adopted technique in text-based recommender systems. Beel et al. (2016) conducted a survey revealing that TF-IDF is the most popular weighting scheme. It combines two components:

- **Term Frequency (TF):** Measures how often a term appears in a document, emphasising frequently used words.
- **Inverse Document Frequency (IDF):** Reduces the weight of terms that are common across many documents, highlighting terms that are more unique to specific documents.

Common Czech stop words and several domain-specific terms were removed prior to computing the TF-IDF matrix; the full list is provided in Table B.1 in Appendix.

The Term Frequency-Inverse Document Frequency (TF-IDF) value for each word is computed as:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \log \left(\frac{N}{\text{DF}(t)} \right),$$

where t is the term, d is the document, N is the total number of documents in the corpus, and $\text{DF}(t)$ is the number of documents that contain the term t .

This formula ensures that terms frequently appearing in a specific document receive a higher weight (TF), while terms that are common across the entire corpus receive a lower weight due to their diminished uniqueness (IDF). As a result, TF-IDF highlights words that are important within a document.

3.5.2 Word2Vec

Word2Vec is a widely used technique for transforming words into dense vector representations in a continuous high-dimensional space, introduced by Mikolov et al. (2013). Unlike traditional bag-of-words or TF-IDF models, which treat words as independent tokens and produce sparse representations, Word2Vec captures semantic and syntactic relationships between words based on their contextual co-occurrence in large corpora.

The core idea behind Word2Vec is the *distributional hypothesis*, which posits that words appearing in similar contexts tend to have similar meanings. Word2Vec operationalises this through two main neural architectures: continuous bag of words (CBOW) (Continuous Bag-of-Words) and *skip-gram*. The CBOW model predicts a target word w_t given its surrounding context $\{w_{t-n}, \dots, w_{t-1}, w_{t+1}, \dots, w_{t+n}\}$, while the skip-gram model does the inverse—it predicts surrounding context words given a central word.

For the skip-gram model, the objective is to maximise the average log probability:

$$\frac{1}{T} \sum_{t=1}^T \sum_{-n \leq j \leq n, j \neq 0} \log P(w_{t+j} | w_t),$$

where T is the total number of words in the corpus, and n is the context window size. The conditional probability is typically computed using the softmax function:

$$P(w_O | w_I) = \frac{\exp((\mathbf{v}'_{w_O})^\top \mathbf{v}_{w_I})}{\sum_{w=1}^V \exp((\mathbf{v}'_w)^\top \mathbf{v}_{w_I})},$$

where $\mathbf{v}_{w_I}$ and $\mathbf{v}'_{w_O}$ are the input and output vector representations of the input word w_I and the context word w_O , respectively, and V is the vocabulary size.

In this thesis, pre-trained Czech word embeddings provided by Kyjánek and Bonami (2022) were used. These embeddings were generated using the CBOW architecture applied to the SYN v9 corpus (Křen et al., 2021), a large linguistic dataset representing contemporary Czech. The choice of CBOW was motivated by empirical findings indicating superior performance for Czech text in comparison with skip-gram (Hořeňovská, 2019; Svoboda and Bryhcín, 2018). The embeddings were 300-dimensional, providing a balance between expressiveness and computational efficiency.

These dense vector representations were then averaged across all words in a property description to yield a fixed-length vector for each listing. These vectors were subsequently integrated as input features into the machine learning models used for real estate price prediction.

3.5.3 BERT

Bidirectional Encoder Representations from Transformers (BERT) is a transformer-based model developed by Devlin et al. (2019) that has set new benchmarks

in natural language understanding tasks. Unlike traditional word embedding methods such as TF-IDF or Word2Vec, which generate context-independent representations, BERT produces dynamic, context-sensitive embeddings by considering both the left and right context of each word through a bidirectional training mechanism.

BERT is based on the Transformer architecture introduced by Devlin et al. (2019), which employs a multi-layer self-attention mechanism to process entire input sequences in parallel. Each input sentence is first tokenised into subword units using a WordPiece tokenizer, resulting in a fixed-length sequence of tokens.

The model is pre-trained using two self-supervised learning objectives:

1. **Masked Language Modeling (MLM):** Random tokens in the input are masked, and the model is trained to predict the original tokens. This enables bidirectional context learning. The loss for a single masked token t is calculated using the cross-entropy between the true token distribution and the model’s softmax output over the vocabulary:

$$\mathcal{L}_{\text{MLM}} = - \sum_{i \in \mathcal{M}} \log P(w_i | \mathbf{x}_{\setminus i}),$$

where $\mathcal{M}$ is the set of masked positions, and $\mathbf{x}_{\setminus i}$ denotes the input sequence with token w_i masked.

2. **Next Sentence Prediction (NSP):** The model is trained to predict whether a given sentence B follows sentence A in the original text. This encourages an understanding of inter-sentence relationships.

In this thesis, the Czech pre-trained model `ufal/robeczech-base` was employed, available through the Hugging Face `transformers` library. This model is based on the BERT-base architecture, consisting of 12 transformer layers (encoder blocks), 12 attention heads, and a hidden size of 768. The model was trained on large-scale Czech corpora, making it particularly well suited for handling property descriptions in Czech.

Each property description was tokenised using `AutoTokenizer` and then passed through the BERT encoder. From the resulting hidden states, the embedding corresponding to the special `[CLS]` token was extracted:

$$\mathbf{h}_{[\text{CLS}]} \in \mathbb{R}^{768},$$

which serves as a fixed-length, sentence-level representation of the input text.

These sentence-level embeddings were computed for all property descriptions and stored in a structured format indexed by property identifiers. They were subsequently integrated with structured features such as location, area, or amenities, and served as input to machine learning models used for real estate price prediction.

The adoption of BERT in this context enables the inclusion of nuanced linguistic information, capturing both syntactic structure and semantic content, thus providing a richer representation of textual data than traditional embedding methods.

3.6 Urban vs Rural Analysis

To assess the differential impact of textual information on predictive performance in urban and rural areas, the dataset is divided into two subsets based on municipality population:

- **Urban Areas:** Municipalities with a population greater than or equal to 10,000 inhabitants.
- **Rural Areas:** Municipalities with a population below 10,000 inhabitants.

This classification ensures a clear distinction between urban and rural areas, enabling an analysis of how model performance varies between these subsets.

The threshold of 10,000 inhabitants for distinguishing urban from rural areas in this study is based on several considerations. First, it aligns conceptually with the European statistical classification, which defines urban areas as municipalities with at least 5,000 inhabitants and a population density of no less than 300 inhabitants per square kilometre (Dijkstra and Poelman, 2014). However, in the Czech Republic, only approximately one third of municipalities with 5,000 to 10,000 inhabitants meet this density criterion. Second, municipalities with over 10,000 inhabitants account for more than 56% Czech residents and total share of real estate offers in these municipalities is even higher (85% in my dataset). Therefore, to achieve a clearer distinction between urban and rural areas, this study adopts a simplified classification by setting the threshold at 10,000 inhabitants.

The following steps are employed to evaluate model performance in urban and rural areas. Firstly, models are trained using only structured data (e.g.,

property size, location, and socio-economic characteristics). Performance metrics, including Mean Absolute Error (MAE)s, Max-Min Mean Absolute Percentage Error (mmMAPE), Log Root Mean Squared Error (LRMSE), and R-Squared (R^2), are computed separately for the urban and rural subsets. Secondly, models are retrained using a combination of structured and unstructured data. Textual information is integrated using the methods described in Section 3.5. Performance metrics are again computed for the urban and rural subsets.

Chapter 4

Data

This chapter introduces my primary dataset of 1,359,000+ Czech apartment listings (2014–2024) obtained from Deloitte and outlines the cleaning and enrichment steps. It begins with raw data description (Section 4.1), then defines my candidate structured features—structural, locational, socio-economic, and temporal (Section 4.2). Next, I extract keywords from listing texts (Section 4.3). Section 4.3 also details Czech-specific text preprocessing for TF-IDF and embeddings.

4.1 Data Description

The data for this thesis comprises structured and unstructured information originally collected from various Czech real estate advertisement portals, covering the period from the 1st of January 2014 to the 31st of December 2024. Figure 4.1 provides a summary of the dataset distribution across the years. Figure 4.2 illustrates the geographical distribution of the listings across the Czech Republic, highlighting the concentration of listings in urban areas.

In this thesis, asking prices from real estate advertisement platforms were used instead of transaction prices. Transaction prices, which refer to the final amounts stated in purchase contracts and collected by the Real estate cadastre, do not provide as much detailed information about the unit. For example, transaction data lack information about the presence of amenities, advertisement duration, and ownership type, which are essential for this study. Furthermore, it is possible to match transactions to less than ten percent of the advertisements, which would significantly limit the dataset’s size and applicability.

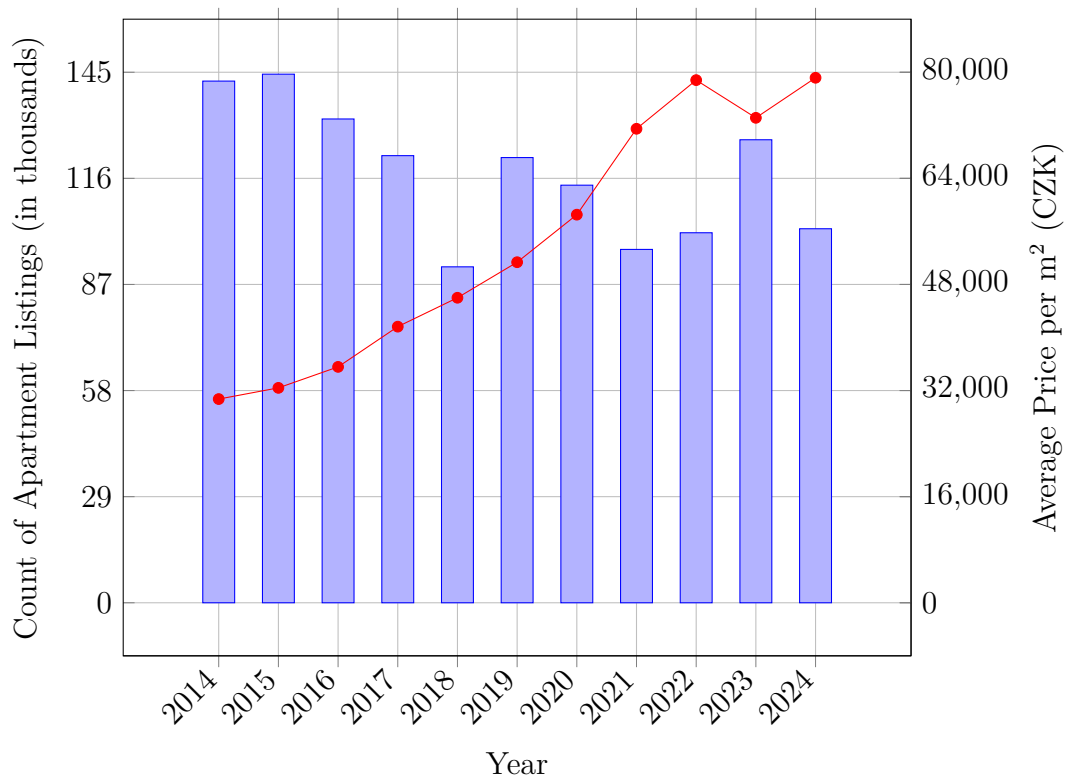


Figure 4.1: Number of apartment listings and average price per m² per year in the Czech Republic

Note: This figure shows the data after all preprocessing steps, including outlier removal.

Therefore, asking price data are used because they offer more comprehensive details about the properties. However, I acknowledge that asking prices serve only as a starting point for negotiation and can differ significantly from agreed transaction prices depending on the phase of the economic cycle, buyer sentiment, and the resulting negotiating power of buyers. Moreover, I do not have information about whether the advertised apartments were ultimately sold. Consequently, the dataset may include properties that did not convert into completed transactions, introducing a potential source of bias.

Multiple features were extracted from the raw data, including structured attributes (e.g., size, number of rooms, amenities) and unstructured text descriptions. The dataset was enriched with locational, socio-economic, and temporal features to enhance the predictive power of the models. The full list of features with a brief description is provided in Table A.2 in Appendix, while their descriptive statistics are summarised in Table A.3 in Appendix.

Deloitte, who provided the data, applied an initial filter based on price per m² (see Table A.1 in Appendix). As a result, more than 1.3 million property

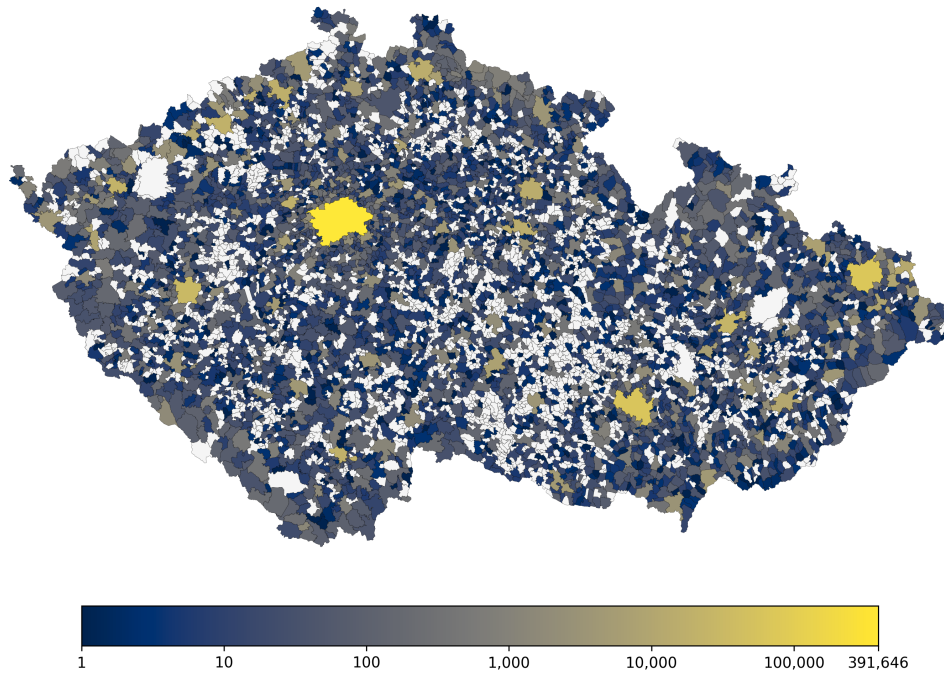


Figure 4.2: Apartment listings in municipalities across the Czech Republic

Note: Based on the final dataset after all filtering steps.

listings were supplied. However, Deloitte did not disclose the detailed methodology used in the data collection process. Therefore, while the price filtering criteria are known, the specific portals or sources from which the data were scraped remain unidentified.

Additionally, I cleansed the dataset by removing missing values in essential columns - `latitude`, `longitude`, `price_per_sqm`, `usable_area`, `rooms`, `condition`, and `building_type`. Furthermore, a series of targeted steps were applied to handle remaining missing entries and extreme values:

- **Ownership.** The `ownership` field, when missing, was imputed to "private".
- **Floor level.** Values ≤ 0 or missing were reset to 1 (ground floor), and listings with `floornumber` > 35 or equal to -1 were removed.
- **Advertisement length.** Negative `advertisement_length` entries were discarded, as were listings advertised for more than two years.

- **IQR-based outlier filtering.** For both `price_sqm` and `usable_area`, the 10th percentile Q_1 and the 90th percentile Q_3 were calculated, then observations outside

$$\left[Q_1 - 1.5 (Q_3 - Q_1), Q_3 + 1.5 (Q_3 - Q_1) \right].$$

were removed.

- **Minimum size threshold.** Finally, any apartment with `usable_area` below 15 m^2 was dropped as likely misreported or non-standard.

These steps ensured the final dataset contains only listings with complete, plausible structural and spatial attributes as well as a well-behaved target variable (`price_sqm`), ready for subsequent feature engineering and modeling. In total, the final dataset consists of 1,295,599 listings. Out of these, 1,076,473 listings are from urban areas (municipalities with a population greater than 10,000), while 219,126 listings are from rural areas (municipalities with a population of 10,000 or less).

4.2 Structured Data

The structured data includes numerical and categorical attributes such as property size (in square meters), number of rooms and presence of amenities (e.g., balconies, elevators). These features are further categorised based on their structural, locational, socio-economic and temporal characteristics.

4.2.1 Structural Features

Structural features describe the intrinsic attributes of a property:

- **Property size (e.g., square footage):** Larger properties are generally associated with lower prices per m^2 .
- **Presence of balcony, kitchen, and elevator:** Features like balconies and elevators significantly enhance the property's utility and attractiveness, especially in urban settings (Baur, Rosenfelder, and Lutz, 2023; Zhang, Li, and Branco, 2024).
- **Apartment condition (e.g., after renovation, before renovation, very good):** The condition of the apartment significantly influences

its valuation, with renovated or well-maintained properties commanding higher prices (Baur, Rosenfelder, and Lutz, 2023; Zhang, Li, and Branco, 2024).

- **Apartment disposition (e.g., studio, 1-bedroom, 2-bedroom):** The layout and functional distribution of living space are important factors influencing the desirability and pricing of apartments (Baur, Rosenfelder, and Lutz, 2023; Zhang, Li, and Branco, 2024).
- **Building type (e.g., panel house, development project, brick house):** It plays a significant role in determining the property's market value (Baur, Rosenfelder, and Lutz, 2023; Zhang, Li, and Branco, 2024).
- **With or without fit out:** Apartments sold with a completed fit out (e.g., flooring, kitchen installation) are usually priced higher than those sold without it (Baur, Rosenfelder, and Lutz, 2023).

4.2.2 Locational and Socio-Economic Features

The integration of spatial and proximity features was motivated by the need to capture locational determinants that are crucial in understanding property price variations. Literature review shows that spatial context, including proximity features, significantly influences property valuations (Selim, 2009; Zhang, Li, and Branco, 2024).

The dataset incorporates GPS coordinates. The accuracy of these data points is limited by the manner in which they are reported. All information in the advertisements is entered by property owners or real estate agents. In some instances, listings are provided with their location specified only at the street level. In such cases, the GPS coordinates represent the midpoint of the street. Even though the precision of these coordinates is not high enough to reliably match all apartment listings to specific buildings in the Czech Real estate cadastre, it allows me to add more locational features.

All point-to-point distances employed haversine-based nearest-neighbour searches, while distances to line and polygon features (e.g. highways, parks) were calculated using R-tree spatial indexing for efficiency. These locational features provide both macro- and micro-level spatial context for the subsequent valuation models.

Shapefile data from the RÚIAN database (ČÚZK, 2012) were used to assign each property to its municipality (see prefix *mun_* in Table A.3), and region

(see prefix *reg_* in Table A.3). If the municipality is divided into districts, they have also been added (see prefix *md_* in Table A.3). These categorical variables reflect differing legal and infrastructural contexts across administrative units. However, most area-specific features were dropped in the VarianceThreshold step, as they did not contribute significantly to the model performance. It is further discussed in Section 4.5.

Using the Data50 topographic model (scale 1:50,000), haversine-based nearest-neighbour distances were computed from each listing to:

- Transport hubs (bus stops, train stations, and metro entrances),
- Administrative centres (county and regional capitals),
- Historical and cultural sites (castles, palaces, churches, synagogues, cemeteries),
- Infrastructure nodes (airports and hospitals),
- Sports facilities and industrial sites (stadiums and tailings dumps),
- Natural features (rivers, water bodies, forests, meadows, national parks),
- Road network elements (highways, including planned segments, and ropeways, especially within mountain resorts).

Given that the influence of distance from the nearest metro station is most likely relevant only for offers within Prague, the inverse value of distance was interacted with the Prague dummy variable (`mun_554782_dist_metro_log`).

To complement the dataset with the distance from the municipal centre, a method based on building entrances was employed. The dataset used, ČSÚ (2015), provided geolocated points for building entrances. The municipal centre was determined as a weighted average of the coordinates of all buildings within the municipality's boundaries. The weights were based on the number of persons, including foreign nationals, recorded in each building as of 31 December 2011. This approach ensured that the centre accurately represented the population distribution within the municipality. Using these weighted coordinates, the distance of each property from the computed municipal centre was calculated.

Beyond the raw variables, several interaction terms and polynomial transformations were created to capture nonlinear effects and tested for their significance (see Section 4.5). Examples include the interaction between municipal population and distance to the city centre (`pop_tot_log_dist_city_cen_`

log), and `dist_city_center_log_2`. The ropeway elevation feature `dist_ropeway_log_max_elevat` combines ropeway elevation with proximity to the nearest ropeway station to capture effect of mountain resorts and their altitudes on property prices.

Regarding socio-economic features, each listing was supplemented with the total population of its municipality. Urban areas generally command higher prices, whereas rural locations tend to exhibit lower valuations (Schulz and Werwatz, 2004; Zhang, Li, and Branco, 2024). In addition, monthly macroeconomic indicators were downloaded from the Czech National Bank’s ARAD API (ČNB, 2002) and merged with the listings. Although these indicators were later removed by the `VarianceThreshold` step due to their limited contribution to model performance, the following variables were initially considered:

- Core inflation,
- Headline inflation,
- the 2T repo rate,
- the product of core inflation and the 2T repo rate,
- the discount rate,
- the average APR on mortgages provided by banks and building societies,
- the general unemployment rate,
- nominal wages.

4.2.3 Temporal Features

Temporal factors play a crucial role in capturing dynamic market conditions and listing behaviours. From the raw timestamp fields (`created_on` and `lastfound_on`), I derived the following variables:

- **Listing duration:** Properties listed for extended periods may indicate overpricing or low demand (Fan, Ong, and Koh, 2006).
- **Market cycles and seasonality:** To control for macroeconomic and seasonal effects, we convert each listing’s `lastfound_on` date into *Year-quarter* dummy variables (2014_Q1, 2014_Q2, etc.) and *Weekday* dummy variables (Monday-Sunday). These features capture quarterly economic

cycles and weekly listing patterns (Baur, Rosenfelder, and Lutz, 2023; Schulz and Werwatz, 2004).

- **Russian invasion to Ukraine and Covid-19 pandemics:** I introduce two binary dummies to account for major demand-shifting events:

$$\begin{aligned} \text{created_after_pandemic} &= \begin{cases} 1 & \text{if created_on} \geq 2020-03-11, \\ 0 & \text{otherwise,} \end{cases} \\ \text{created_after_invasion} &= \begin{cases} 1 & \text{if created_on} \geq 2022-02-24, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

These indicators allow my model to adjust for structural shifts in buyer sentiment and market liquidity induced by the Covid-19 pandemic and the Russian invasion of Ukraine.

4.3 Unstructured Data

4.3.1 Keyword-based Extractions of Features

The textual descriptions and titles of property listings were analysed to extract specific features, focusing on the presence of lofts and gardens. Two new binary variables, `loft` and `garden`, were derived using keyword-based searches and contextual filtering to ensure accuracy.

The identification of lofts relied on occurrences of the word *loft* in the `description` and `title` fields. To minimise false positives, the following steps were applied:

- Listings containing the keyword *loft* were flagged.
- Exclusion criteria were applied to filter out mentions of unrelated terms, such as *mezonet* or *penthouse*.

The identification of gardens combined textual and structural information:

- Listings containing the keyword *zahrada* in the `description` or `title` fields were flagged. The declension of the keyword was taken into account.
- Only properties located on the ground floor or first floor were considered eligible for classification.

- Exclusion rules were implemented to avoid false positives, removing cases where terms like *shared garden* or *view to garden* indicated shared or non-accessible gardens.

4.3.2 Textual Data Preprocessing for NLP techniques

Natural Language Processing (NLP) methods suggested in Section 3.5 need preprocessing steps. To ensure compatibility with Czech property descriptions, specific preprocessing steps were designed to address linguistic features unique to the Czech language.

Data Preprocessing for TF-IDF

TF-IDF is frequency-based and performs best with clean, stop-word-free text. Therefore, the following preprocessing steps were applied. Mostly the steps were implemented using TF-IDF Vectorizer in the Sklearn package (Pedregosa et al., 2011):

1. **Numbers:** All numbers were normalised. Ordinal and cardinal numbers in text (e.g., *jedna* or *první*) were converted to numeric format (e.g., 1, 1.).
2. **Abbreviations and Acronyms:** Common abbreviations in Czech property descriptions (e.g., *m²* for square metres, *kk* for kitchenette) were standardised to a single consistent form. A manual inspection of a sample dataset was conducted to identify and normalise these variations.
3. **Typos and Misspellings:** Detected errors were corrected using Czech-language dictionaries.
4. **Punctuation and Special Characters:** All punctuation marks and special characters were removed, except for +, which was replaced by the word *plus*.
5. **Lemmatisation:** Czech-specific lemmatisation was applied using the MorphoDiTa library (Straka and Straková, 2014), which effectively handles the inflectional morphology of the Czech language.
6. **Stop Words:** Both common Czech stop words (e.g., *a*, *v*, *na*) and real estate-specific terms (e.g., *lokalita*, *metr*, *plocha*) were removed, as they provided minimal predictive value (see Table B.1 in Appendix).

7. **Tokenisation:** Text was tokenised. The words were ignored if they appeared in less than 3% of all the descriptions.

The 3% cut-off was chosen empirically to filter out extremely rare words that appear only in a handful of listings. In the Python preprocessing script, this is implemented via `TfidfVectorizer(min_df=0.03)`, ensuring that only terms occurring in at least three percent of the descriptions contribute to the feature space. This choice reduces noise and keeps the dimensionality of the TF-IDF matrix manageable while retaining the most informative vocabulary.

Data Preprocessing for Word2Vec

The preprocessing steps for Word2Vec were mostly identical to those used for TF-IDF, with one important exception:

1. **Retention of Stop Words:** Unlike TF-IDF, Word2Vec benefits from contextual information provided by frequently occurring words. Therefore, stop words were retained to preserve sentence structure and context.

Data Preprocessing for BERT

The preprocessing for BERT-based embeddings diverged significantly from the approaches used for TF-IDF and Word2Vec, owing to the requirements of transformer-based models. Specifically, the Czech language model `ufal/robeczech-base` was employed through the Hugging Face `transformers` library. The preprocessing pipeline focused on ensuring compatibility with the model's tokenisation and input constraints.

1. **Tokenizer-Based Normalisation:** Instead of manually applying lemmatisation or stop-word removal, input texts were passed directly to the `AutoTokenizer` associated with the pre-trained Czech BERT model. The tokenizer handles token segmentation into subword units using a Word-Piece vocabulary tailored to the Czech language.
2. **Padding and Truncation:** Descriptions were padded and truncated to a maximum sequence length of 512 tokens. This limit corresponds to the maximum context size that BERT models can process. Any text exceeding this length was truncated, ensuring compliance with the model architecture.

3. **Special Tokens:** The tokenizer automatically appended special tokens required for the transformer input format.
4. **Vector Extraction:** After encoding, each property description was passed to the `robeczech-base` model in inference mode. From the resulting hidden state tensor, the vector representation corresponding to the `[CLS]` token (i.e., the first token in the sequence) was extracted:

$$\mathbf{h}_{[\text{CLS}]} \in \mathbb{R}^{768},$$

providing a fixed-length embedding summarising the entire input text.

Unlike frequency-based methods, BERT does not require explicit removal of punctuation, numbers, or stop words. Instead, these elements are handled implicitly within the tokenisation and attention mechanisms of the model, making additional linguistic preprocessing unnecessary.

4.4 Text Feature Reduction Analysis

Given the high dimensionality introduced by categorical variables, particularly those representing administrative units such as municipalities, districts, and regions, dimensionality reduction was applied to mitigate multicollinearity and reduce the risk of overfitting. These categorical variables were encoded as binary (dummy) features, resulting in a sparse feature space with a large number of indicators, many of which corresponded to municipalities with very few listings.

To address this, the `VarianceThreshold` method, a univariate feature selection technique, was employed. This method removes features whose variance does not exceed a specified threshold across the dataset. In the context of binary variables, this effectively removes those that appear in only a small fraction of observations—i.e., municipalities that are underrepresented in the dataset. Such features contribute minimal discriminative power and can introduce noise and instability in model estimation.

The threshold was set to 0.01, meaning that a dummy variable had to be present in at least 1% of all listings to be retained. This mirrors the setting `VarianceThreshold(threshold=0.01)` in the preprocessing code and prunes municipalities with just a few occurrences from the feature space.

By filtering out low-variance dummy variables, the dimensionality of the regional representation was significantly reduced. This step improved computational efficiency, enhanced model generalisability, and prioritised geographical features with broader representation and greater explanatory value across the sample.

Furthermore, dimensionality reduction was applied to the textual embeddings generated in the preprocessing scripts. Averaged Word2Vec and BERT vectors were projected onto their first 100 principal components using Principal Component Analysis (PCA). The sparse TF-IDF matrix was handled with truncated singular value decomposition via `TruncatedSVD` and the same component count. Regional dummies were not subjected to PCA; they were only filtered with the `VarianceThreshold` step. Retaining only these components preserved most of the variance while keeping the resulting feature matrices compact enough for the subsequent machine-learning models.

4.5 Multicollinearity Reduction by Feature Selection

During preprocessing, a large set of categorical variables was created by one-hot encoding regional identifiers—region (`VUSC_KOD`), municipality (`OBEC_KOD`), and municipal-district codes (`KOD_MC`). The resulting design matrix contained thousands of sparse binary columns and exhibited pronounced multicollinearity, especially between nested spatial hierarchies. The accompanying condition number exceeded 10^{14} , which rendered the Lasso estimator numerically unstable.

Pairwise Pearson correlations were therefore computed for all numeric and one-hot-encoded variables. Whenever the absolute correlation coefficient exceeded 0.9, the variable in each pair with the lower mutual information (MI) with respect to the target price was removed.

Each modeling pipeline was subsequently estimated twice—once with the full feature set and once with the filtered set—in an A/B test. For Lasso, Ridge, and Random Forest, retaining the highly correlated variables significantly increased mean absolute error (MAE) and lowered R^2 . In contrast, XGBoost benefited from the additional features. Consequently, the correlated variables were kept only for the XGBoost specification and were removed when fitting the Lasso, Ridge, and Random Forest models.

The removed features comprised all region and municipal-district dummies together with several squared and interaction terms:

- `usable_area_log_2`
- `dist_city_center_log_2`
- `pop_tot_log_dist_city_cen_log_2`
- `created_after_invasion_created_after_pandemic`
- `created_after_invasion_population_total_log`
- `created_after_pandemic_population_total_log`

Removing these highly collinear variables reduced the condition number of the design matrix and improved the stability of the Lasso, Ridge, and Random Forest estimates.

Chapter 5

Results

5.1 Lasso and Ridge Regression

The baseline Lasso model serves as a benchmark for understanding the effect of each structured predictor. Because the dependent variable is the log of price per m^2 , coefficients can be interpreted approximately as percentage changes in price. Table D.1 in Appendix lists all coefficients for the Lasso model. Variables related to location dominate. For the basic scenario, population size, usable area, distance to the nearest regional centre, distance to the city centre, and the presence of a separate kitchen or private ownership status appear with the largest absolute values. A one percent increase in population size, for example, reduces price per m^2 by roughly 0.13% when the interaction with distance to the city centre is ignored. This negative coefficient is merely an artefact of the model parametrisation; once the interaction is accounted for, higher municipal population raises price per m^2 almost everywhere except right at the centroid, fully in line with urban-economics theory and the wider empirical literature. Larger usable area is associated with a lower price per m^2 , reflecting the common discount for spacious flats. Having a private ownership right raises the price by about 9%. These magnitudes accord with prior findings that urban amenities and market liquidity play a decisive role in Czech housing prices (Baur, Rosenfelder, and Lutz, 2023; A. Nowak and P. Smith, 2017).

Introducing textual features leaves the sign of these coefficients unchanged and only reduces their magnitude, indicating that embeddings explain part of the variance captured by structured variables. The ranking of key predictors remains stable across TF-IDF, Word2Vec, and BERT representations. Figures D.1 to D.4 in Appendix show the coefficient trajectories.

Ridge regression provides a complementary perspective by shrinking all coefficients toward zero without eliminating any predictors (see Table D.2 in Appendix). The five variables with the largest absolute effects mirror those highlighted by the Lasso model: population size, usable area, private ownership, distance to the closest regional centre and the presence of a separate kitchen. Coefficients are smaller in magnitude because of the ℓ_2 penalty, yet the signs are consistent. Larger municipalities and private ownership correspond to higher prices, whereas greater floor space and remoteness from regional hubs reduce price per square metre. Adding textual embeddings causes only minor changes in magnitude and preserves the overall ordering of predictors across TF-IDF, Word2Vec, and BERT. Figures D.5 to D.8 in Appendix summarise coefficient paths.

5.2 Random Forest

The Random Forest ranks features by their influence on prediction error (for more detail see Table E.1 in Appendix). The dummy indicating that a listing was created after the COVID-19 outbreak is the single most important variable, highlighting the large market shift that occurred during the pandemic. Population size of the municipality is the next strongest factor, confirming that larger urban areas command higher prices. The remaining top features measure accessibility—distance to the nearest ropeway with high elevation, distance to the nearest airport and distance to the closest synagogue, a proxy for historical town centres. Their strong impact underlines how proximity to transportation and amenities boosts prices. This ordering is consistent with the linear models where population and distance variables also exhibit the largest coefficients. Incorporating textual features slightly redistributes importance but the same variables remain at the top, implying that structured data already capture the bulk of the predictive signal.

As shown in Figure E.1 in Appendix, the training and validation curves exhibit a noticeable gap that narrows only gradually with more training data. This behaviour indicates that the Random Forest is variance limited.

5.3 XGBoost

For XGBoost the ranking of the most influential predictors closely mirrors the Random Forest results. Table E.2 in Appendix summarises variable importance for XGBoost. The variable marking the post-pandemic period again tops the list, followed by population size and measures of accessibility such as distance to ropeways, the nearest airport and national parks. Usable area also remains among the strongest effects. These factors emphasise the dominance of broader location characteristics over detailed building attributes. Their relative importance is consistent with the coefficient magnitudes observed in the linear models, suggesting that tree-based methods extract similar high-level signals despite their non-linear nature.

Gain-based importance additionally highlights several interaction terms involving these variables. The inclusion of textual embeddings produces only modest shifts in the relative rankings, underscoring that structured features carry most of the predictive weight.

Figure E.2 in Appendix shows near-overlapping training and validation curves that flatten quickly. This pattern indicates a bias-limited model for which additional observations offer little improvement.

5.4 Model Performance Comparison

The overall test set evaluation reveals clear differences in predictive accuracy across the four examined models: Lasso, Ridge, Random Forest, and XGBoost. Table 5.1 summarises the predictive performance of those four models when trained on structured data only and when augmented with TF-IDF, Word2Vec or BERT text features. The linear models—Lasso and Ridge regression—perform nearly identically, with Lasso marginally outperforming Ridge. Without embeddings Lasso achieves a mean absolute error (Mean Absolute Error (MAE)) of CZK 11,639 and a coefficient of determination of R-Squared (R^2) = 0.779, while Ridge yields an MAE of CZK 11,645 and R^2 = 0.778. These results indicate a limited capacity of linear models to capture complex non-linear interactions in the housing data.

In contrast, the ensemble methods exhibit substantially improved predictive performance. Random Forest achieves the best results across all evaluated metrics, with the lowest MAE of CZK 5,255 and the highest R^2 value of 0.943. It also demonstrates superior performance in logarithmic root mean squared

error (Log Root Mean Squared Error (LRMSE) = 0.187) and modified mean absolute percentage error (Max-Min Mean Absolute Percentage Error (mmMAPE) = 0.144), suggesting robust generalisation across the dataset. XGBoost, while slightly less accurate than Random Forest, still significantly outperforms the linear models. It records an MAE of CZK 7,690 and an R^2 of 0.893, further confirming the advantage of ensemble-based methods for housing price prediction.

These findings support a growing body of literature that demonstrates the superiority of machine learning algorithms-particularly ensemble approaches-over traditional regression techniques in real estate valuation tasks (Baur, Rosenfelder, and Lutz, 2023; Guo et al., 2020; Park and Bae, 2015). The ability of Random Forest and XGBoost to model complex, non-linear dependencies allows for more precise price estimates, making them highly suitable for practical implementation in automated valuation models. Table 5.1 summarises these results.

Table 5.1: Comparison of overall model performance using structured features only, TF-IDF, Word2Vec, and BERT text features.

Model	Feature Type	MAE	mmMAPE	LRMSE	R^2
Lasso	No Text	11,639	0.344	0.353	0.779
	TF-IDF	11,187	0.314	0.329	0.792
	Word2Vec	11,153	0.320	0.330	0.795
	BERT	11,102	0.316	0.328	0.797
Ridge	No Text	11,645	0.344	0.353	0.778
	TF-IDF	11,202	0.313	0.329	0.791
	Word2Vec	11,177	0.320	0.330	0.794
	BERT	11,117	0.316	0.328	0.796
Random Forest	No Text	5,255	0.144	0.187	0.943
	TF-IDF	4,871	0.133	0.179	0.948
	Word2Vec	5,103	0.139	0.184	0.944
	BERT	5,373	0.145	0.189	0.939
XGBoost	No Text	7,690	0.202	0.236	0.893
	TF-IDF	7,394	0.192	0.226	0.901
	Word2Vec	5,952	0.149	0.183	0.934
	BERT	5,979	0.150	0.184	0.934

Among linear models, both Lasso and Ridge show clear improvements with textual data. For Lasso, the MAE drops from CZK 11,639 without text to CZK 11,187 with TF-IDF and CZK 11,153 with Word2Vec, while R^2 increases

from 0.779 to 0.792 and 0.795. Ridge exhibits a similar pattern, with MAE decreasing from CZK 11,645 to CZK 11,202 (TF-IDF) and CZK 11,177 (Word2Vec) and R^2 rising from 0.778 to 0.791 and 0.794. BERT embeddings provide the largest gains, reducing MAE to CZK 11,102 for Lasso and CZK 11,117 for Ridge with R^2 values of 0.797 and 0.796. These gains are accompanied by modest reductions in mmMAPE and log-RMSE.

For ensemble models, the effect of textual features is mixed. Random Forest gains slightly from TF-IDF (MAE 4,871 compared with 5,255) and shows little change with Word2Vec, while BERT leads to a small decline. By contrast, XGBoost improves noticeably: MAE drops from 7,690 to 7,394 with TF-IDF (about a 4% reduction) and further to 5,952 with Word2Vec; BERT achieves a similar 5,979. These results suggest that while linear models benefit substantially from textual information—especially in capturing variance that structured features alone may not explain—ensemble models may already extract sufficient signal from structured data or require more advanced integration techniques for unstructured inputs.

5.5 Model Performance Difference between Rural and Urban Areas

The performance comparison of the four predictive models on the test dataset—segmented by municipality size—highlights important differences in model effectiveness and generalisability. Table 5.2 compares these approaches for all four scenarios across regions. Both linear models exhibit notable discrepancies in prediction accuracy between municipalities with populations above and below 10,000 inhabitants. For municipalities over 10,000, Lasso achieves an MAE of CZK 11,448 and $R^2 = 0.799$, while Ridge attains CZK 11,458 and the same R^2 . Performance deteriorates substantially in smaller municipalities, with Lasso reaching an MAE of CZK 12,576 and $R^2 = 0.525$, and Ridge MAE = CZK 12,563 with $R^2 = 0.528$. The increased error and reduced R^2 in smaller municipalities suggest that linear models struggle to capture the greater heterogeneity and noise characteristic of rural housing markets.

Ensemble models deliver markedly superior performance across all test segments. The Random Forest model attains the lowest overall MAE (CZK 5,255), with a high R^2 of 0.943, indicating a strong ability to capture non-linear relationships. Notably, Random Forest performs consistently well across munic-

ipality types. In municipalities over 10,000 inhabitants, it achieves a MAE of CZK 5,361 and an R^2 of 0.944, while in smaller municipalities it still performs strongly with a MAE of CZK 4,732 and $R^2 = 0.916$. Similarly, XGBoost maintains robust predictive accuracy, though slightly below that of Random Forest, with a test MAE of CZK 7,722 and $R^2 = 0.898$ for larger municipalities, and CZK 7,533 and $R^2 = 0.819$ for smaller ones.

A more nuanced view is offered by the modified mean absolute percentage error (mmMAPE), which normalises the relative error across price scales. Linear models again show significant deterioration in mmMAPE for smaller municipalities (e.g., 0.544 for Lasso and 0.545 for Ridge) compared to larger ones (0.304 and 0.303, respectively). In contrast, Random Forest achieves mmMAPE values of 0.136 and 0.182 for larger and smaller municipalities, respectively, suggesting greater adaptability to rural market complexity. The superior generalisability of Random Forest can likely be attributed to its inherent capacity to model interactions among diverse features and capture spatial and structural heterogeneity, which is more pronounced in rural contexts (Baur, Rosenfelder, and Lutz, 2023; Guo et al., 2020; Park and Bae, 2015).

These results underscore the advantages of ensemble methods, particularly Random Forest, in generating robust property valuations across varying demographic and spatial contexts. The pronounced performance gap of linear models between urban and rural municipalities reflects their limited flexibility in capturing complex, region-specific pricing dynamics. Ensemble models, by contrast, demonstrate both higher accuracy and resilience across heterogeneous environments, aligning with findings by Guo et al. (2020) and Zhang, Li, and Branco (2024) that machine learning approaches integrating contextual variability outperform traditional regression-based methods. Detailed results by municipality size are listed in Table 5.2.

Table 5.2: Comparison of model performance with structured data only, TF-IDF, Word2Vec, and BERT features for urban and rural subsets.

Model	Subset	MAE	mmMAPE	LRMSE	R^2
Lasso					
No Text	Urban	11,448	0.304	0.322	0.799
	Rural	12,576	0.544	0.479	0.525
TF-IDF	Urban	11,153	0.284	0.303	0.807
	Rural	11,352	0.460	0.435	0.603
Word2Vec	Urban	11,099	0.285	0.304	0.810
	Rural	11,416	0.494	0.435	0.608
BERT	Urban	11,071	0.284	0.304	0.811
	Rural	11,254	0.475	0.429	0.617
Ridge					
No Text	Urban	11,458	0.303	0.321	0.799
	Rural	12,563	0.545	0.479	0.528
TF-IDF	Urban	11,178	0.283	0.303	0.805
	Rural	11,317	0.460	0.434	0.608
Word2Vec	Urban	11,137	0.284	0.304	0.807
	Rural	11,374	0.495	0.434	0.612
BERT	Urban	11,093	0.284	0.303	0.809
	Rural	11,235	0.477	0.429	0.620
Random Forest					
No Text	Urban	5,361	0.136	0.177	0.944
	Rural	4,732	0.182	0.228	0.916
TF-IDF	Urban	4,944	0.125	0.168	0.949
	Rural	4,514	0.174	0.223	0.920
Word2Vec	Urban	5,194	0.131	0.173	0.945
	Rural	4,654	0.179	0.227	0.917
BERT	Urban	5,462	0.137	0.178	0.941
	Rural	4,936	0.189	0.235	0.910
XGBoost					
No Text	Urban	7,722	0.187	0.221	0.898
	Rural	7,533	0.277	0.301	0.819
TF-IDF	Urban	7,402	0.177	0.210	0.906
	Rural	7,359	0.266	0.290	0.833
Word2Vec	Urban	6,064	0.141	0.174	0.935
	Rural	5,402	0.190	0.223	0.908
BERT	Urban	6,081	0.141	0.175	0.935
	Rural	5,478	0.192	0.225	0.904

The inclusion of unstructured text via TF-IDF features consistently improves the performance of linear models, especially in rural areas where the structured data are often sparse. For instance, Ridge regression’s rural mmMAPE decreases from 0.545 to 0.460 when TF-IDF features are added, and the corresponding R^2 rises from 0.528 to 0.608. Word2Vec embeddings provide further marginal improvements in rural subsets but are less effective in urban settings. BERT embeddings follow the same pattern, offering marginally higher accuracy than Word2Vec, particularly in rural municipalities. Ensemble models such as Random Forest and XGBoost show no significant gains from text-based features, suggesting that these algorithms already capture the nonlinear relationships sufficiently from structured variables alone.

5.6 Comparison with Previous Research

The results obtained in this thesis confirm that the inclusion of unstructured textual features improves the predictive performance of linear models such as Lasso and Ridge regression. Specifically, the addition of TF-IDF features led to a decrease in Mean Absolute Error (MAE) of roughly 4% for both Lasso and Ridge, while Word2Vec embeddings yielded comparable improvements. The marginal mean absolute percentage error (mmMAPE) was reduced by around 15% in rural areas, with R^2 values rising from about 0.78 to 0.79–0.80 overall and to roughly 0.61 for rural subsets. BERT embeddings yielded the best overall accuracy, although the gains over Word2Vec were small.

These findings align with recent literature. Baur, Rosenfelder, and Lutz (2023) reported up to a 17.1% reduction in MAE when property descriptions were integrated via vector representations, particularly for low-priced listings. Similarly, Zhang, Li, and Branco (2024) observed up to an 8% improvement in RMSE when including unstructured text, with stronger gains in rural or low-information settings. A. Nowak and P. Smith (2017) also found that real estate agent descriptions reduced pricing error by more than 25% using penalised Lasso, although their tokenised MLS data may have been of higher quality than general listing texts.

For non-linear models such as Random Forest and XGBoost, no significant performance gains were observed with the addition of TF-IDF, Word2Vec or BERT features. In several cases, performance metrics worsened slightly, suggesting redundancy or overfitting. This outcome is consistent with the findings of Guo et al. (2020), who also observed limited to no improvements for tree-based

models when text embeddings were added. Stevens (2014) similarly reported only modest text-related gains in regression tasks, despite using rich n-gram feature sets.

Overall these findings mirror the broader consensus in recent literature. An important contribution of this study is the explicit comparison of model performance between urban and rural municipalities, which shows that textual information is particularly helpful when structured data are limited. Linear models profit markedly from textual descriptions, particularly where structured data are sparse as in rural municipalities. TF-IDF and Word2Vec embeddings offer comparable gains, although the simpler TF-IDF representation is often slightly superior. By contrast, ensemble models such as Random Forest and XGBoost show little improvement from text features unless additional feature engineering or model adaptations are employed.

Chapter 6

Discussion

The empirical analysis confirms that ensemble models outperform linear regression, yet textual features are most beneficial for the latter. Particularly in rural municipalities, the addition of apartment descriptions helps compensate for the limited availability of structured data. These findings echo earlier studies that reported the usefulness of unstructured text for improving price estimates (Baur, Rosenfelder, and Lutz, 2023; Zhang, Li, and Branco, 2024).

Even though unstructured data improve model performance, optimal dimensionality of BERT, TF-IDF and Word2Vec was not investigated in depth. The number of principal components was chosen heuristically, leaving room for further experimentation. Future work could refine the dimensionality reduction step—for instance by cross-validating the number of PCA components or employing alternative techniques such as autoencoders—to assess whether more informative embedding representations translate into better predictions.

Another open question concerns the quality of the textual descriptions themselves. Anecdotal evidence suggests that real estate listings are increasingly drafted with generative tools such as ChatGPT. If this trend continues, descriptions may become more homogeneous, potentially reducing the predictive power of text-based features. Evaluating the impact of automatically generated descriptions on model performance warrants further investigation once more data become available.

In addition to richer text, future models could also incorporate visual information. Photographs of the apartments and their surroundings are readily accessible on real estate websites and might capture aspects of quality not fully reflected in the structured attributes. Combining image-based features with the existing textual and numerical data offers a direction for extending the

presented approach.

From a practical perspective, the results are directly relevant for professionals involved in property valuation. The strong performance of Random Forest and XGBoost demonstrates that automated valuation models can achieve high accuracy, especially when sufficient structured data are available. However, wider use of the models, for example for mass appraisal, is hampered by their challenging interpretability. In the case of non-linear models, the effects of individual features on the price per m² cannot be easily separated. In addition, the mechanics of the models themselves are highly complex and thus not very transparent, which makes it difficult to be checked and/or even approved by the institution in charge of financial market supervision, in the case of the Czech Republic the Czech National Bank. Linear models have a better starting position in this respect, but have a significantly lower performance. The results show that this performance can be compensated by including text embeddings. However, the text embeddings themselves, with the exception of TF-IDF, cannot be easily interpreted, which brings us to the same issue as in the case of non-linear models. Thus, the results point to a way to greatly improve the accuracy of the estimates in rural areas, but the adoption of these models poses a major challenge for real estate practitioners and the institution in charge of supervising the financial market.

Chapter 7

Conclusion

This thesis set out to evaluate whether textual descriptions from real estate listings can enhance the prediction of apartment prices across urban and rural contexts. To this end, a dataset of more than 1.3 million Czech property listings (2014–2024) was compiled and enriched with locational, socio-economic, and temporal features. Unstructured text was converted to numerical form using TF-IDF, Word2Vec, and BERT embeddings, providing complementary information to structured attributes such as size, amenities, and location.

Four predictive models were compared: Ridge and Lasso regression on the linear side, and Random Forest and XGBoost among ensemble methods. The empirical results demonstrated a clear advantage of ensemble learning. Random Forest achieved the lowest MAE of CZK 5,255 and an R^2 of 0.943, with XGBoost following closely. Linear models showed inferior predictive performance but benefited substantially from the inclusion of textual data, particularly in rural municipalities where structured information is scarce.

Textual embeddings proved most valuable when integrated with regression models. Word2Vec and BERT features reduced MAE by around 2–5% relative to purely structured baselines, and the gains were greatest in rural areas. Ensemble methods, however, extracted little additional signal from the text, suggesting that their inherent ability to capture complex interactions already subsumes much of the information contained in property descriptions.

The findings underscore two key contributions. First, they highlight the predictive potential of a large-scale dataset spanning the entire Czech market, including regions rarely examined in prior research. Second, they show that simple, interpretable models can approach the accuracy of more complex algorithms when augmented with well-engineered textual features. This insight is

of practical value for institutions that require transparent valuation tools yet operate in data-sparse environments.

Future work could extend the analysis by refining the dimensionality reduction of text embeddings, exploring non-linear architectures tailored to combined structured and unstructured inputs, and incorporating image data from listings. Such enhancements may further narrow the performance gap between linear and ensemble methods while preserving interpretability.

Overall, this thesis demonstrates that while modern ensemble algorithms remain the strongest performers for mass appraisal, textual descriptions provide a meaningful boost to linear models and thus deserve consideration in any comprehensive valuation workflow.

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Appendix A

Detailed Data Description

Table A.1: Basic Price Filters Applied by Deloitte

Year	Min Apartment Sale Value (CZK)	Max Apartment Sale Value (CZK)
2014	2,000	192,500
2015	2,300	209,000
2016	2,600	225,500
2017	3,100	242,000
2018	3,400	258,500
2019	3,700	275,000
2020	4,000	275,000
2021	4,300	295,000
2022	5,000	350,000
2023	7,500	350,000
2024	8,500	400,000

Note: Data provided by Deloitte.

Table A.2: Descriptions of model variables

Feature	Description
price_sqm_log	Log price per m ² in CZK
advertisement_length_log	Log of listing duration in days
balcony	Indicator for balcony
barrier_free	Indicator for barrier free
bldg_mixed	Building type: mixed
bldg_prefabricated	Building type: prefabricated
bldg_skeleton	Building type: skeleton
cond_before refurbishment	Condition: before refurbishment
cond_good	Condition: good
cond_new development	Condition: new development
cond_project	Condition: project
cond_under construction	Condition: under construction
cond_very good	Condition: very good
created_after_invasion	Listing created after Russian invasion of Ukraine (24th Feb 2022)
created_after_invasion_created_after_pandemic	Interaction: pandemic dummy $\times$ invasion dummy
created_after_invasion_population_total_log	Interaction: invasion dummy $\times$ log population
created_after_pandemic	Listing created after the first Czech lockdown due to Covid-19 pandemic (11th Mar 2020)
created_after_pandemic_population_total_log	Interaction: pandemic dummy $\times$ log population
dist_airport_log	Log distance to airport
dist_castle_log	Log distance to castle
dist_chateau_log	Log distance to chateau
dist_church_log	Log distance to church
dist_city_center_log	Log distance to city center
dist_city_center_log_2	Squared log distance to municipality centre
dist_closest_county_center_log	Log distance to closest county center
dist_closest_regional_center_log	Log distance to closest regional center
dist_forest_log	Log distance to forest
dist_highway_log	Log distance to highway
dist_hospital_log	Log distance to hospital
dist_meadow_log	Log distance to meadow
dist_np_log	Log distance to the closest national park
dist_ropeway_log_max_elevat	Log distance to the closest ropeway * elevation of the highest point served by the ropeway
dist_stadium_log	Log distance to stadium
dist_synagogue_log	Log distance to synagogue
dist_tailings_dump_log	Log distance to tailings dump
dist_water_area_log	Log distance to water area

Continued on next page

Feature	Description
dist_water_flow_log	Log distance to water flow
elevator	Indicator for elevator
equipped	Indicator for equipped
floornumber	Floor number
floornumber_elevator	Floor number multiplied by indicator of elevator
garage	Indicator for garage
garden	Indicator for garden
loggia	Indicator for loggia
own_private	Ownership: private
parking_space	Indicator for parking space
pop_tot_log_dist_city_cen_log	Log population $\times$ log distance to municipality centre
pop_tot_log_dist_city_cen_log_2	Squared interaction of log population and log distance
population_total_log	Log population of municipality
rooms	Number of rooms
sep_kitchen	Indicator for sep kitchen
terrace	Indicator for terrace
usable_area_log	Log usable area
usable_area_log_2	Squared log usable area
md_	Dummy variables based on municipality district code according to the RÚIAN codebook.
mun_	Dummy variables based on municipality code according to the RÚIAN codebook.
weekday_	Dummy variables representing the day of the week.
year_quarter_	Dummy variables based on listing year and quarter.
reg_	Dummy variables based on region code according to the RÚIAN codebook.

Table A.3: Summary statistics of all model variables

Feature	Unique	Mean	SD	Min	Max
price_sqm_log	163,733	10.59	0.80	7.60	12.42
advertisement_length_log	730	3.51	1.52	0.69	6.59
balcony	2	0.09	0.28	0.00	1.00
barrier_free	2	0.29	0.45	0.00	1.00
bldg_mixed	2	0.06	0.25	0.00	1.00
bldg_prefabricated	2	0.41	0.49	0.00	1.00
bldg_skeleton	2	0.03	0.16	0.00	1.00
cond_before refurbishment	2	0.03	0.18	0.00	1.00
cond_good	2	0.23	0.42	0.00	1.00
cond_new development	2	0.15	0.36	0.00	1.00
cond_project	2	0.01	0.11	0.00	1.00
cond_under construction	2	0.03	0.17	0.00	1.00
cond_very good	2	0.38	0.49	0.00	1.00
created_after_invasion	2	0.23	0.42	0.00	1.00

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Feature	Unique	Mean	SD	Min	Max
created_after_invasion_created_after_pandemic	2	0.23	0.42	0.00	1.00
created_after_invasion_population_total_log	1,784	2.58	4.85	0.00	14.10
created_after_pandemic	2	0.38	0.49	0.00	1.00
created_after_pandemic_population_total_log	1,886	4.33	5.68	0.00	14.10
dist_airport_log	347,740	9.74	0.87	4.79	11.65
dist_castle_log	347,740	9.08	0.92	2.70	11.05
dist_chateau_log	347,740	8.42	0.99	1.66	10.54
dist_church_log	347,740	6.39	0.89	0.85	9.25
dist_city_center_log	347,740	7.21	1.01	0.78	9.05
dist_city_center_log_2	347,740	53.04	14.11	0.60	81.82
dist_closest_county_center_log	347,740	8.17	1.15	0.78	10.66
dist_closest_regional_center_log	347,740	9.01	1.27	1.71	11.41
dist_forest_log	308,341	5.65	2.11	0.00	8.14
dist_highway_log	347,740	8.99	0.96	0.39	11.39
dist_hospital_log	347,740	7.56	1.06	2.33	10.64
dist_meadow_log	336,013	6.19	1.38	0.00	8.37
dist_np_log	347,733	11.66	0.65	0.00	12.72
dist_ropeway_log_max_elevat	347,740	82.37	13.75	60.91	1,920
dist_stadium_log	347,740	7.51	1.01	2.26	10.30
dist_synagogue_log	347,740	8.94	1.32	2.33	11.28
dist_tailings_dump_log	347,740	8.66	0.78	3.85	11.03
dist_water_area_log	336,536	6.75	1.54	0.00	9.61
dist_water_flow_log	347,740	6.27	1.08	0.01	8.63
elevator	2	0.21	0.41	0.00	1.00
equipped	2	0.13	0.33	0.00	1.00
floornumber	34	3.10	2.31	1.00	34.00
floornumber_elevator	26	0.82	1.98	0.00	33.00
garage	2	0.22	0.42	0.00	1.00
garden	2	0.04	0.19	0.00	1.00
loggia	2	0.42	0.49	0.00	1.00
own_private	2	0.82	0.38	0.00	1.00
parking_space	2	0.08	0.27	0.00	1.00
pop_tot_log_dist_city_cen_log	347,744	1.59	0.23	0.59	13.29
pop_tot_log_dist_city_cen_log_2	347,744	2.57	0.81	0.35	176.66
population_total_log	2,000	11.45	2.24	3.81	14.10
price_sqm	163,771	52,657	37,833	2,000	246,562
rooms	7	2.39	0.89	1.00	7.00
sep_kitchen	2	0.55	0.50	0.00	1.00
terrace	2	0.25	0.43	0.00	1.00
usable_area_log	172	4.13	0.37	2.77	5.23
usable_area_log_2	172	17.20	3.01	7.69	27.36

Note: Mean for booleans is the share of TRUE; textual columns report the mode and its frequency. Dummy variables for regions, municipalities and municipality districts were used in the models but are not displayed in this table. Number of observations after all preprocessing steps, including outlier removal: 1,295,599.

Appendix B

Textual Data

Table B.1: Stop words removed prior to TF-IDF processing

ahoj	do	kdy	nebyla	plocha	tobě
ale	dobrý	když	nebyli	plzeň	tohle
anebo	docela	ke	nebyly	po	toto
ano	dva	kolik	nedělají	potom	trošku
asi	dvacet	kromě	nedělá	pozdě	tvoje
aspoň	dvanáct	která	nedělám	pořád	tvá
atd	dvě	které	neděláme	praha	tvé
ačkoli	dál	který	neděláte	pro	tvůj
bez	dále	kterí	neděláš	prostě	ty
beze	děkovat	kvůli	nejsi	prosím	tě
blízko	děkujeme	kč	nemají	proti	třeba
bohužel	děkuji	liberec	nemáme	protože	tři
brno	dům	lokalita	nemáte	proč	třináct
brzo	hodně	mají	neměl	pět	určitě
bude	jak	metr	není	před	už
budeme	jde	mezi	nestačí	přes	vary
budete	je	mnou	nevadí	přese	vaše
budeš	jeden	mně	než	rovně	vaši
budou	jedenáct	moc	nic	se	ve
budu	jedna	mohl	nich	sedm	vedle

Continued on next page

budějovice	jedno	mohou	nimi	sedmnáct	velikost
byl	jednou	moje	nula	skoro	večer
byla	jedou	moji	nám	smí	vlastně
byli	jeho	možná	námi	smějí	vy
bylo	jejich	musí	nás	snad	vám
byly	její	my	náš	spolu	vámi
bys	jemu	má	ním	sta	vás
byt	jen	málo	ně	sto	váš
být	jenom	mám	něco	sté	všechno
během	jestli	máme	nějak	ta	všichni
chce	jestliže	máte	někde	tady	vůbec
chceme	ještě	máš	někdo	tak	vždy
chcete	jich	mé	němu	takhle	za
chceš	jimi	mí	od	taky	zatímco
chci	jinak	mít	ode	tam	zač
chtít	jsem	mě	olomouc	tamhle	ze
chtějí	jsi	můj	on	tamhleto	zlín
chut'	jsme	může	ona	tamto	čau
chuti	jsou	na	oni	tebe	český
cirka	jste	nad	ono	tebou	čtrnáct
co	já	nade	ony	ted'	čtyři
cz	jí	naproti	osm	tedy	šest
daleko	jím	naše	osmnáct	ten	šestnáct
den	kam	naši	ostrava	ti	že
deset	karlovy	ne	pak	tisíc	
devatenáct	kde	nebo	pardubice	tisíce	
devět	kdo	nebyl	patnáct	to	

Appendix C

Grid Search for Hyperparameter Optimisation

Table C.1: Model hyperparameters and their description

Model	Parameter	Description
Lasso	alpha	Regularisation strength (L1 for Lasso, L2 for Ridge). A higher alpha increases penalty on large coefficients, encouraging simpler models and reducing overfitting.
Ridge	alpha	Regularisation strength (L1 for Lasso, L2 for Ridge). A higher alpha increases penalty on large coefficients, encouraging simpler models and reducing overfitting.
RF	n_estimators	Number of trees in the ensemble. Increasing this often raises predictive accuracy but adds computational cost and risk of overfitting.
	max_depth	Maximum depth of each tree. Lower values constrain model complexity and help generalise, while higher values allow capturing deeper interactions.
	max_features	Number or fraction of features to consider at each split. Lower values introduce more randomness (decorrelation), while higher values may improve split quality but risk overfitting.
XGBoost	n_estimators	Number of trees in the ensemble. Increasing this often raises predictive accuracy but adds computational cost and risk of overfitting.
	max_depth	Maximum depth of each tree. Lower values constrain model complexity and help generalise, while higher values allow capturing deeper interactions.
	learning_rate	Boosting step size (shrinkage). Lower values slow learning, requiring more trees but often producing more robust models.

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Model	Parameter	Description
	subsample	Fraction of training instances to use for each tree. Values <1.0 introduce stochasticity, improving generalisation at the cost of slight bias.
	colsample_bytree	Fraction of features to sample when constructing each tree. Reduces feature correlation and overfitting; lower values increase variance reduction.
	gamma	Minimum loss reduction (gain) required to split a leaf. Higher gamma makes the algorithm more conservative, avoiding splits with low information gain.
	min_child_weight	Minimum sum of instance weights (Hessian) needed in a child node. Higher values prevent trees from creating nodes with few samples, reducing overfitting.
	reg_alpha	L1 regularisation on weights. Adds sparsity and can improve generalisation in tree ensembles.
	reg_lambda	L2 regularisation on weights. Helps control model complexity and avoid overfitting.

Table C.2: Hyperparameter search grid and final values

Model	Parameter	Grid	Scenario	Value
Lasso	alpha	Stage 1: 15 values logarithmically spaced between 0.00001 and 1; Stage 2: $\log_{10}(\text{best}) \pm 0.5$; max_iter=[20k, 50k]; tol=1e-4	A	0.0003049
			B	0.0003938
			C	0.0003938
			D	0.0003049
Ridge	alpha	Stage 1: 40 values log-spaced between 1e-6 and 1e6 with solvers auto, sag, lsqr; Stage 2: $\log_{10}(\text{best}) \pm 1$ (plus 0 if solver is not sag)	A	1.343
			B	13.43
			C	25.72
			D	25.72
RF	n_estimators	Stage 1: [200, 500, 800]; Stage 2: elbow $\pm$ 100, 200	A	800
			B	100
			C	100
			D	100
	max_depth	Stage 1: [None, 10]	A	N/A
			B	N/A
			C	N/A
			D	N/A
XGBoost	n_estimators	Stage 1: [0.3, sqrt, 0.5]	A	sqrt
			B	0.3
			C	0.3
			D	0.3
	n_estimators	Stage 1: [100, 300, 500, 800, 1000, 1200, 1500, 2000]; Stage 2: elbow $\pm$ 100, 200	A	1800
			B	1800
			C	1300

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Model	Parameter	Grid	Scenario	Value
	max_depth	Stage 1: [3, 5, 7]	D	1300
			A	3
			B	3
			C	7
	learning_rate	Stage 1: [0.05, 0.1, 0.2]	D	7
			A	0.05
			B	0.05
			C	0.05
	subsample	Stage 1: [0.6, 0.8, 1.0]	D	0.05
			A	0.6
			B	0.6
			C	1
	colsample_bytree	Stage 1: [0.6, 0.8, 1.0]	D	1
			A	0.6
			B	0.6
			C	0.6
	gamma	Stage 1: [0, 0.1]	D	0.6
			A	0
			B	0
			C	0
	min_child_weight	Stage 1: [1, 5, 10]	D	0
			A	1
			B	1
			C	5
	reg_alpha	Stage 1: [0, 0.1, 0.5]	D	5
			A	0
			B	0
			C	0.1
	reg_lambda	Stage 1: [1, 1.5, 2.0]	D	0.1
			A	1
			B	1
			C	1
			D	1

Note: Scenario labels A–D refer to: A = no embeddings; B = TF-IDF; C = Word2Vec; D = BERT. Two-stage hyperparameter tuning was used. Stage 1 ranges are listed first; Stage 2 ranges show the refined search if applicable.

Appendix D

Results of Linear Models

D.1 LASSO Regression

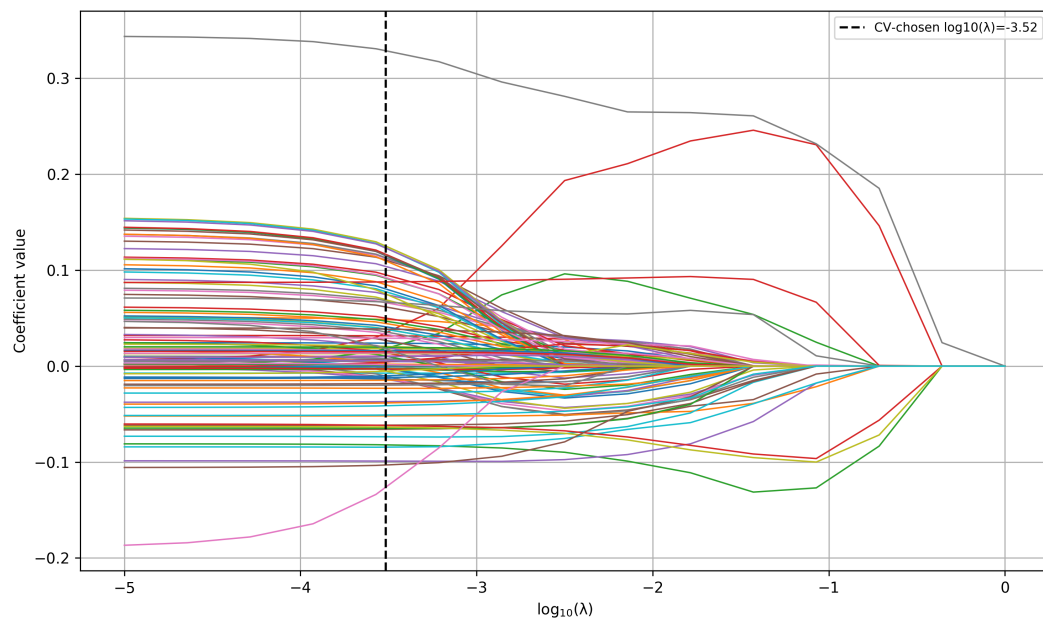


Figure D.1: Lasso coefficient paths (baseline scenario)

Note: Created using the 80% training split.

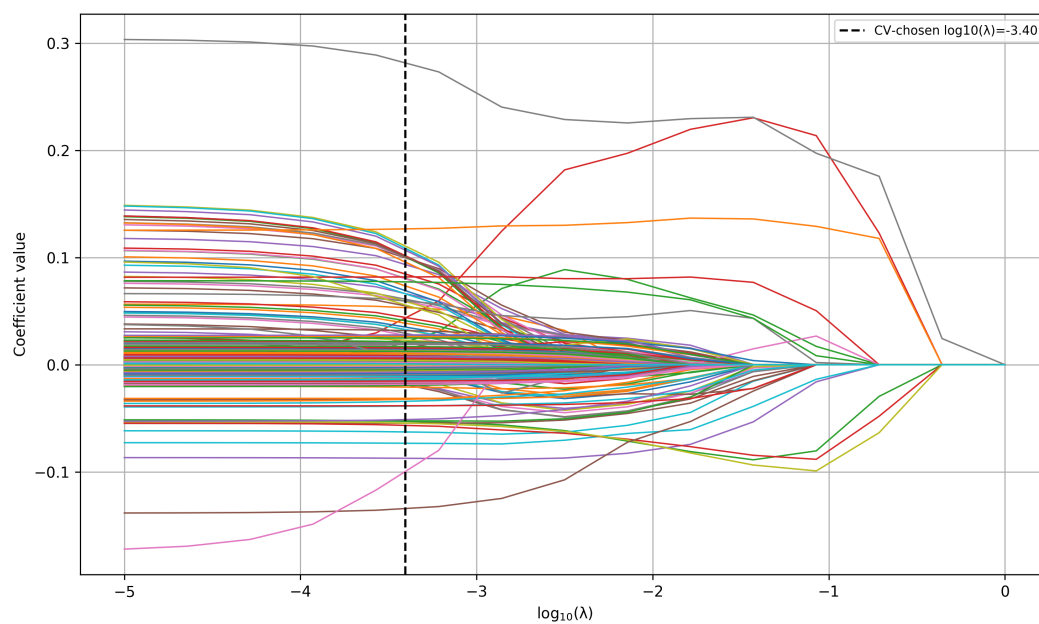


Figure D.2: Lasso coefficient paths (TF-IDF scenario)

Note: Created using the 80% training split.

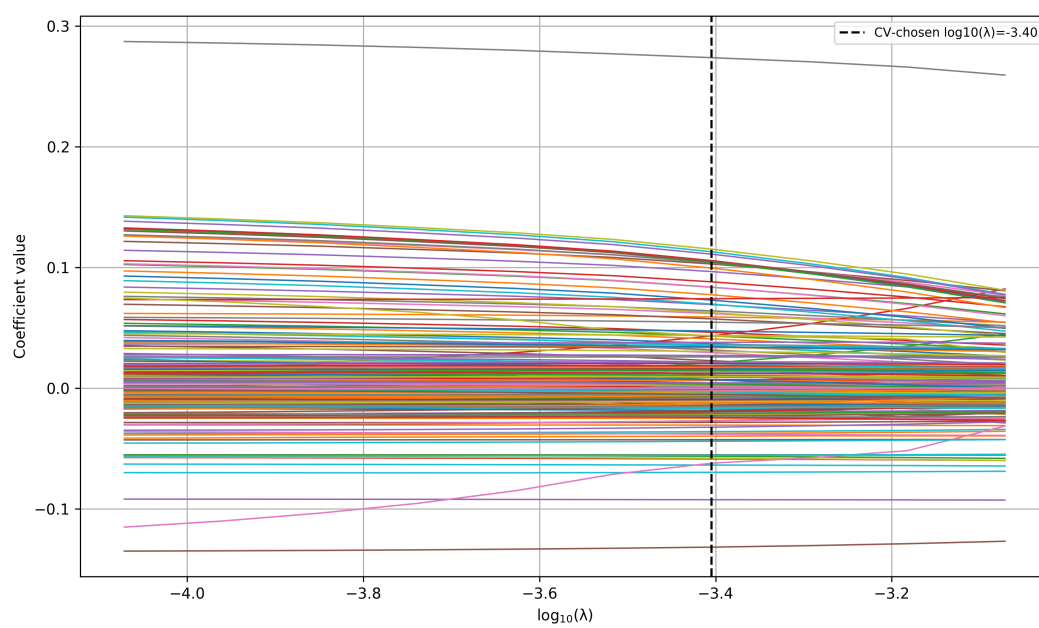


Figure D.3: Lasso coefficient paths (Word2Vec scenario)

Note: Created using the 80% training split.

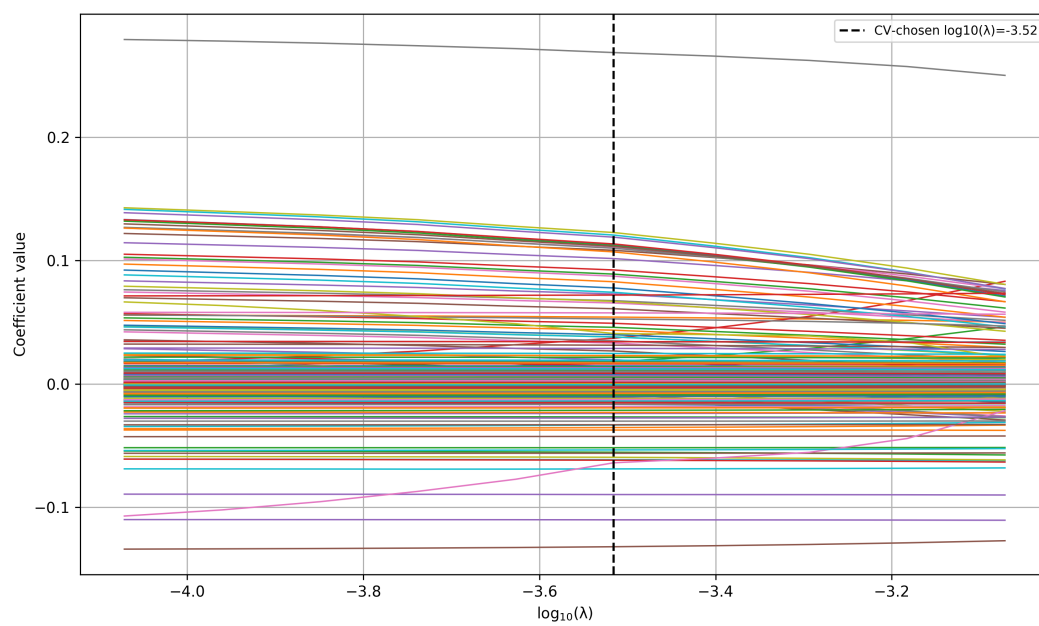


Figure D.4: Lasso coefficient paths (BERT scenario)

Note: Created using the 80% training split.

Table D.1: LASSO coefficients comparison

feature	no embeddings		TF-IDF		Word2Vec		BERT	
	coeff	$\Delta\%$	coeff	$\Delta\%$	coeff	$\Delta\%$	coeff	$\Delta\%$
advertisement_ length_log	0.0041	0.4157	0.0058	0.5847	0.0025	0.2475		
balcony	0.0316	3.2090	0.0166	1.6733	0.0169	1.7047	0.0174	1.7554
barrier_free	0.0182	1.8324	0.0119	1.2013	0.0120	1.2077	0.0122	1.2246
bldg_mixed	-0.0119	-1.1877	-0.0098	-0.9707	-0.0101	-1.0021	-0.0122	-1.2077
bldg_prefabricated	-0.0519	-5.0536	-0.0314	-3.0904	-0.0307	-3.0244	-0.0374	-3.6697
bldg_skeleton	-0.0031	-0.3096	-0.0064	-0.6353	-0.0057	-0.5664	-0.0061	-0.6060
cond_ before refurbishment	-0.0371	-3.6376	-0.0190	-1.8851	-0.0245	-2.4222	-0.0240	-2.3675
cond_good	-0.0658	-6.3717	-0.0321	-3.1577	-0.0428	-4.1863	-0.0426	-4.1667
cond_new development	0.0292	2.9600	0.0122	1.2308	0.0185	1.8668	0.0142	1.4341
cond_project	0.0054	0.5410	0.0004	0.0433	0.0032	0.3173	0.0003	0.0281
cond_ under construction	0.0130	1.3091	0.0054	0.5427	0.0100	1.0040	0.0054	0.5401
cond_very good	-0.0278	-2.7398	-0.0092	-0.9135	-0.0170	-1.6898	-0.0165	-1.6360
dist_airport_log	-0.0618	-5.9932	-0.0566	-5.4991	-0.0587	-5.7018	-0.0618	-5.9960
dist_castle_log	-0.0182	-1.8059	-0.0170	-1.6898	-0.0138	-1.3698	-0.0142	-1.4096
dist_chateau_log	0.0138	1.3846	0.0138	1.3907	0.0137	1.3805	0.0135	1.3574
dist_church_log	-0.0199	-1.9713	-0.0182	-1.8002	-0.0150	-1.4895	-0.0166	-1.6433
dist_city_center_ log	0.0751	7.7966	0.0489	5.0121	0.0432	4.4148	0.0408	4.1657
dist_closest_ county_center_log	-0.0102	-1.0135	-0.0133	-1.3177	-0.0100	-0.9910	-0.0107	-1.0667
dist_closest_ regional_center_log	-0.0846	-8.1087	-0.0734	-7.0732	-0.0698	-6.7437	-0.0689	-6.6554
dist_forest_log	0.0150	1.5064	0.0156	1.5715	0.0158	1.5929	0.0162	1.6307
dist_highway_log	-0.0384	-3.7708	-0.0323	-3.1827	-0.0376	-3.6871	-0.0354	-3.4753
dist_hospital_log	-0.0414	-4.0532	-0.0342	-3.3613	-0.0361	-3.5443	-0.0334	-3.2820
dist_meadow_log	0.0057	0.5756	0.0063	0.6340	0.0070	0.7027	0.0072	0.7210
dist_np_log	-0.0649	-6.2800	-0.0524	-5.1075	-0.0556	-5.4076	-0.0517	-5.0373
dist_ropeway_log_ max_elevat	-0.0510	-4.9727	-0.0386	-3.7898	-0.0557	-5.4174	-0.0535	-5.2108
dist_stadium_log	0.0099	0.9908	0.0071	0.7117	0.0091	0.9126	0.0080	0.8035
dist_synagogue_log	-0.0635	-6.1545	-0.0543	-5.2862	-0.0582	-5.6507	-0.0595	-5.7733
dist_tailings_ dump_log	0.0162	1.6297	0.0147	1.4760	0.0124	1.2436	0.0146	1.4692
dist_water_area_ log	-0.0147	-1.4608	-0.0130	-1.2962	-0.0086	-0.8599	-0.0103	-1.0265
dist_water_flow_ log	0.0196	1.9807	0.0169	1.7067	0.0167	1.6856	0.0178	1.7947
elevator	0.0078	0.7807	0.0005	0.0535			0.0025	0.2515
equipped	0.0067	0.6737	0.0023	0.2334	0.0021	0.2099	0.0032	0.3212
floornumber	0.0016	0.1579	-0.0044	-0.4349	-0.0019	-0.1913	-0.0040	-0.3994
floornumber_elevator	-0.0053	-0.5327	-0.0042	-0.4197	-0.0051	-0.5065	-0.0058	-0.5742
garage	-0.0073	-0.7319	-0.0047	-0.4658	-0.0066	-0.6591	-0.0062	-0.6178
garden	0.0020	0.2027	0.0037	0.3657			0.0005	0.0468
loggia	0.0107	1.0794	0.0033	0.3307	0.0068	0.6800	0.0033	0.3294
own_private	0.0878	9.1722	0.0820	8.5405	0.0740	7.6840	0.0720	7.4633
parking_space	0.0048	0.4785	-0.0030	-0.2980	0.0008	0.0801	0.0007	0.0696
pop_tot_log_dist_ city_cen_log	0.0176	1.7778	0.0002	0.0218				
population_total_ log	-0.1264	-11.8739	-0.0911	-8.7091	-0.0623	-6.0389	-0.0639	-6.1928

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feature	no embeddings		TF-IDF		Word2Vec		BERT	
	coeff	$\Delta\%$	coeff	$\Delta\%$	coeff	$\Delta\%$	coeff	$\Delta\%$
rooms	0.0486	4.9835	0.0536	5.5017	0.0589	6.0651	0.0541	5.5618
sep_kitchen	-0.0820	-7.8771	-0.0528	-5.1421	-0.0566	-5.4991	-0.0553	-5.3839
terrace	0.0239	2.4209	0.0137	1.3842	0.0167	1.6814	0.0123	1.2411
usable_area_log	-0.1032	-9.8037	-0.1346	-12.5932	-0.1318	-12.3478	-0.1319	-12.3606
municipality code:								
500496	0.0004	0.0367	0.0049	0.4929				
532053	0.0386	3.9330	0.0319	3.2434	0.0336	3.4142	0.0314	3.1874
544256	0.0159	1.5985	0.0169	1.7055	0.0172	1.7397	0.0176	1.7760
554782	0.3289	38.9481	0.2822	32.6081	0.2737	31.4857	0.2685	30.7982
554791	0.0205	2.0743	0.0232	2.3459	0.0207	2.0966	0.0227	2.3000
554804	-0.0737	-7.1028	-0.0627	-6.0809	-0.0638	-6.1800	-0.0614	-5.9538
554821	-0.0135	-1.3404	-0.0051	-0.5039	-0.0186	-1.8403	-0.0165	-1.6405
554961	-0.0232	-2.2902	-0.0208	-2.0625	-0.0251	-2.4752	-0.0231	-2.2844
555134	0.0009	0.0897	0.0050	0.5053	0.0027	0.2673	0.0023	0.2309
563889	-0.0030	-0.3039	-0.0012	-0.1244	-0.0015	-0.1497	-0.0017	-0.1723
567027	-0.0990	-9.4237	-0.0874	-8.3693	-0.0925	-8.8348	-0.0896	-8.5744
567442	-0.0616	-5.9703	-0.0546	-5.3157	-0.0553	-5.3833	-0.0562	-5.4622
569810	0.0016	0.1566	0.0064	0.6447	0.0025	0.2538	0.0022	0.2224
582786	0.0667	6.8929	0.0597	6.1570	0.0545	5.6064	0.0526	5.4043
date:								
2014 Q2	-0.0136	-1.3467	-0.0163	-1.6128	-0.0168	-1.6708	-0.0133	-1.3182
2014 Q3	-0.0124	-1.2287	-0.0175	-1.7358	-0.0156	-1.5517	-0.0124	-1.2328
2014 Q4	-0.0125	-1.2471	-0.0168	-1.6707	-0.0145	-1.4352	-0.0109	-1.0878
2015 Q1	-0.0101	-1.0055	-0.0151	-1.4986	-0.0128	-1.2720	-0.0086	-0.8530
2015 Q2	-0.0048	-0.4813	-0.0103	-1.0212	-0.0085	-0.8415	-0.0050	-0.5029
2015 Q3	-0.0016	-0.1611	-0.0056	-0.5629	-0.0041	-0.4089	-0.0015	-0.1452
2015 Q4	-0.0024	-0.2366	-0.0059	-0.5912	-0.0051	-0.5064	-0.0013	-0.1298
2016 Q1	0.0026	0.2645	-0.0005	-0.0478			0.0037	0.3743
2016 Q2	0.0105	1.0518	0.0053	0.5362	0.0070	0.7050	0.0107	1.0728
2016 Q3	0.0141	1.4174	0.0091	0.9188	0.0108	1.0843	0.0142	1.4326
2016 Q4	0.0199	2.0057	0.0148	1.4933	0.0164	1.6526	0.0196	1.9787
2017 Q1	0.0270	2.7362	0.0218	2.2042	0.0227	2.2972	0.0267	2.7056
2017 Q2	0.0335	3.4083	0.0282	2.8614	0.0294	2.9792	0.0331	3.3628
2017 Q3	0.0359	3.6528	0.0303	3.0729	0.0316	3.2082	0.0350	3.5627
2017 Q4	0.0388	3.9520	0.0335	3.4065	0.0348	3.5425	0.0381	3.8850
2018 Q1	0.0386	3.9312	0.0337	3.4321	0.0348	3.5425	0.0380	3.8766
2018 Q2	0.0412	4.2015	0.0360	3.6623	0.0370	3.7685	0.0399	4.0697
2018 Q3	0.0446	4.5627	0.0396	4.0380	0.0407	4.1568	0.0436	4.4514
2018 Q4	0.0471	4.8190	0.0419	4.2779	0.0431	4.4088	0.0458	4.6915
2019 Q1	0.0502	5.1495	0.0449	4.5972	0.0462	4.7294	0.0491	5.0357
2019 Q2	0.0749	7.7829	0.0675	6.9873	0.0691	7.1594	0.0729	7.5622
2019 Q3	0.0619	6.3858	0.0559	5.7482	0.0573	5.8979	0.0609	6.2810
2019 Q4	0.0663	6.8551	0.0602	6.2020	0.0617	6.3662	0.0656	6.7772
2020 Q1	0.0684	7.0826	0.0621	6.4094	0.0640	6.6057	0.0675	6.9811
2020 Q2	0.0697	7.2215	0.0606	6.2436	0.0625	6.4544	0.0668	6.9122
2020 Q3	0.0782	8.1316	0.0679	7.0290	0.0696	7.2079	0.0743	7.7079
2020 Q4	0.0811	8.4430	0.0710	7.3556	0.0729	7.5581	0.0778	8.0858
2021 Q1	0.0855	8.9235	0.0754	7.8279	0.0773	8.0394	0.0828	8.6304
2021 Q2	0.0923	9.6649	0.0823	8.5834	0.0836	8.7148	0.0890	9.3066
2021 Q3	0.0957	10.0376	0.0863	9.0117	0.0880	9.2010	0.0924	9.6750
2021 Q4	0.1045	11.0112	0.0951	9.9806	0.0965	10.1321	0.1016	10.6909
2022 Q1	0.1111	11.7551	0.1015	10.6809	0.1028	10.8261	0.1083	11.4355
2022 Q2	0.1131	11.9704	0.1024	10.7802	0.1039	10.9542	0.1103	11.6610
2022 Q3	0.1138	12.0478	0.1028	10.8218	0.1037	10.9260	0.1100	11.6231
2022 Q4	0.1261	13.4374	0.1134	12.0082	0.1151	12.2021	0.1227	13.0510
2023 Q1	0.1243	13.2339	0.1114	11.7862	0.1129	11.9576	0.1206	12.8192

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feature	no embeddings		TF-IDF		Word2Vec		BERT	
	coeff	$\Delta\%$	coeff	$\Delta\%$	coeff	$\Delta\%$	coeff	$\Delta\%$
2023 Q2	0.1160	12.3018	0.1036	10.9138	0.1051	11.0835	0.1131	11.9767
2023 Q3	0.1107	11.7028	0.0984	10.3365	0.0994	10.4525	0.1070	11.2920
2023 Q4	0.1165	12.3595	0.1033	10.8773	0.1043	10.9893	0.1121	11.8578
2024 Q1	0.1175	12.4722	0.1043	10.9952	0.1056	11.1401	0.1134	12.0114
2024 Q2	0.1240	13.2010	0.1095	11.5722	0.1110	11.7343	0.1189	12.6274
2024 Q3	0.1164	12.3499	0.1033	10.8875	0.1050	11.0748	0.1114	11.7894
2024 Q4	0.0930	9.7440	0.0821	8.5608	0.0837	8.7299	0.0873	9.1234
weekday:								
Monday	-0.0004	-0.0407			-0.0002	-0.0178		
Tuesday	-0.0025	-0.2470	-0.0017	-0.1706	-0.0018	-0.1777	-0.0015	-0.1524
Wednesday					0.0002	0.0152		
Thursday	-0.0008	-0.0769	-0.0011	-0.1109	-0.0008	-0.0823	-0.0012	-0.1175
Friday	0.0017	0.1659	0.0014	0.1363	0.0015	0.1519	0.0017	0.1728
Saturday	-0.0029	-0.2919	-0.0031	-0.3135	-0.0036	-0.3613	-0.0029	-0.2848
created_after__ invasion	0.0175	1.7625	0.0194	1.9611	0.0208	2.0977	0.0179	1.8055
created_after__ pandemic	0.0332	3.3782	0.0407	4.1493	0.0432	4.4102	0.0370	3.7700

Note: Columns show coefficients for each scenario; $\Delta\%$ is the implied percentage change.

D.2 Ridge Regression

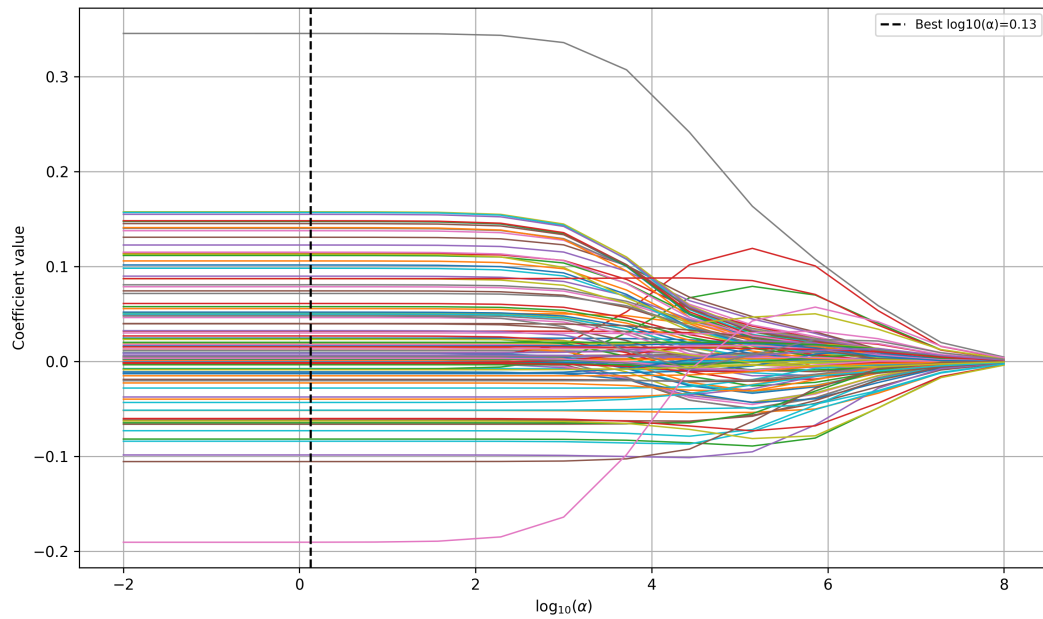


Figure D.5: Ridge coefficient paths (baseline scenario)

Note: Created using the 80% training split.

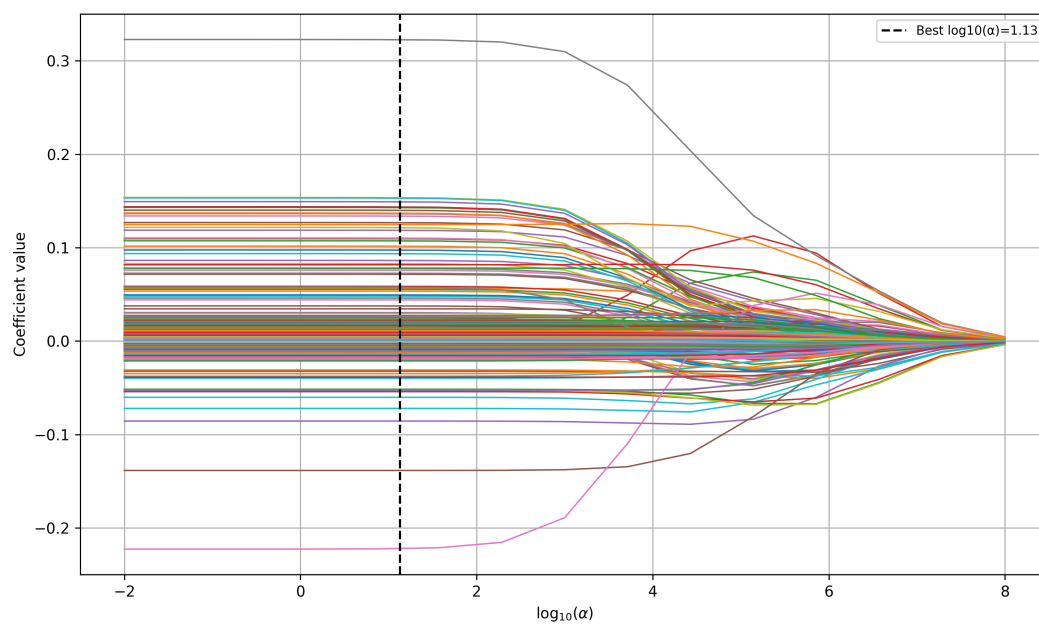


Figure D.6: Ridge coefficient paths (TF-IDF scenario)

Note: Created using the 80% training split.

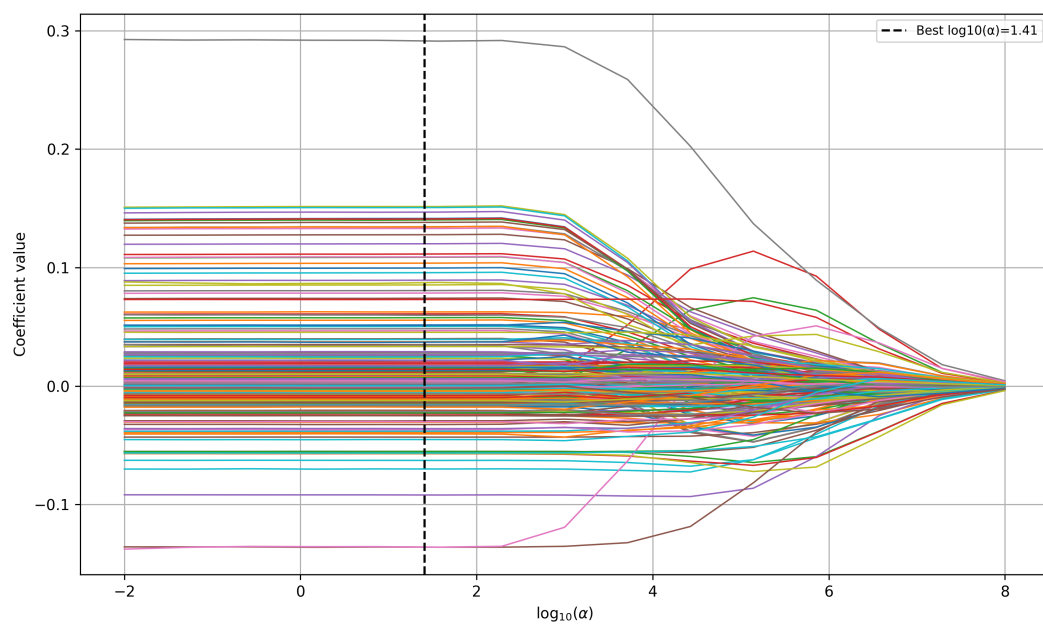


Figure D.7: Ridge coefficient paths (Word2Vec scenario)

Note: Created using the 80% training split.

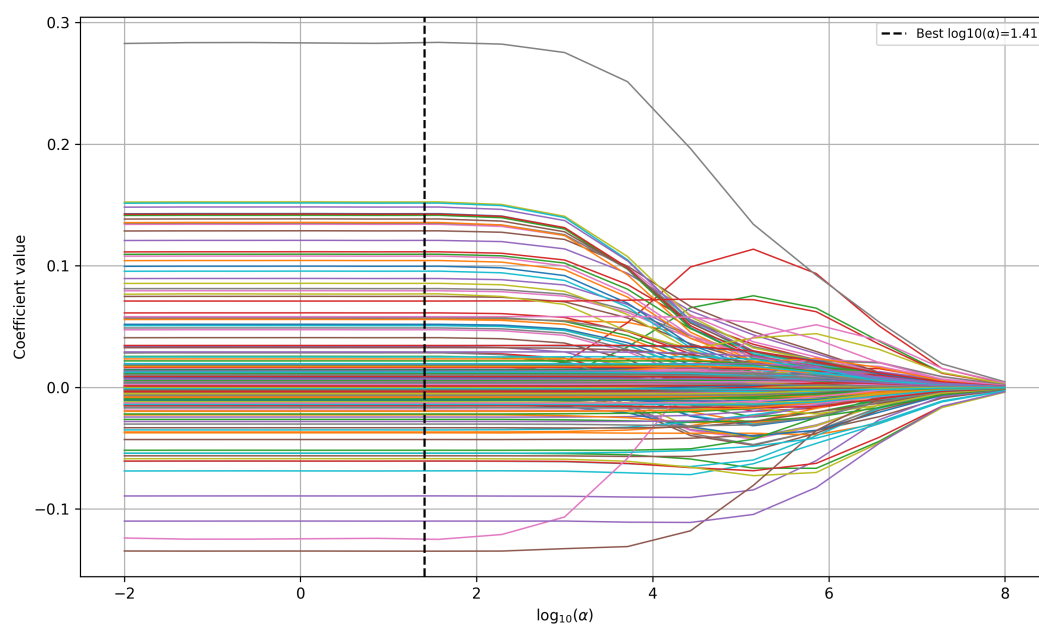


Figure D.8: Ridge coefficient paths (BERT scenario)

Note: Created using the 80% training split.

Table D.2: RIDGE coefficients comparison

feature	no embeddings		TF-IDF		Word2Vec		BERT	
	coeff	$\Delta\%$	coeff	$\Delta\%$	coeff	$\Delta\%$	coeff	$\Delta\%$
advertisement_ length_log	-0.0011	-0.1139	-0.0008	-0.0774	-0.0039	-0.3883	-0.0055	-0.5463
balcony	0.0319	3.2444	0.0165	1.6613	0.0174	1.7563	0.0180	1.8115
barrier_free	0.0186	1.8742	0.0121	1.2198	0.0122	1.2319	0.0122	1.2262
bldg_mixed	-0.0129	-1.2829	-0.0109	-1.0824	-0.0111	-1.1011	-0.0130	-1.2960
bldg_prefabricated	-0.0516	-5.0287	-0.0308	-3.0351	-0.0305	-3.0077	-0.0373	-3.6647
bldg_skeleton	-0.0035	-0.3489	-0.0069	-0.6903	-0.0062	-0.6163	-0.0064	-0.6425
cond_ before refurbishment	-0.0375	-3.6760	-0.0195	-1.9357	-0.0250	-2.4684	-0.0243	-2.4054
cond_good	-0.0662	-6.4017	-0.0323	-3.1802	-0.0431	-4.2183	-0.0429	-4.1966
cond_new development	0.0300	3.0461	0.0136	1.3743	0.0197	1.9855	0.0150	1.5140
cond_project	0.0052	0.5233	0.0006	0.0619	0.0036	0.3635	0.0004	0.0372
cond_ under construction	0.0130	1.3107	0.0057	0.5669	0.0105	1.0538	0.0055	0.5530
cond_very good	-0.0280	-2.7657	-0.0093	-0.9292	-0.0172	-1.7071	-0.0166	-1.6444
dist_airport_log	-0.0601	-5.8354	-0.0536	-5.2198	-0.0571	-5.5544	-0.0605	-5.8733
dist_castle_log	-0.0189	-1.8709	-0.0173	-1.7182	-0.0144	-1.4271	-0.0147	-1.4586
dist_chateau_log	0.0133	1.3368	0.0129	1.2975	0.0133	1.3400	0.0132	1.3333
dist_church_log	-0.0203	-2.0119	-0.0185	-1.8345	-0.0152	-1.5122	-0.0169	-1.6769
dist_city_center_ log	0.1131	11.9741	0.1215	12.9136	0.0866	9.0434	0.0767	7.9755
dist_closest_ county_center_log	-0.0126	-1.2484	-0.0184	-1.8271	-0.0127	-1.2617	-0.0129	-1.2786
dist_closest_ regional_center_log	-0.0841	-8.0629	-0.0720	-6.9505	-0.0700	-6.7589	-0.0686	-6.6318
dist_forest_log	0.0154	1.5513	0.0164	1.6520	0.0162	1.6350	0.0165	1.6668
dist_highway_log	-0.0397	-3.8950	-0.0348	-3.4247	-0.0392	-3.8404	-0.0366	-3.5962
dist_hospital_log	-0.0432	-4.2299	-0.0376	-3.6880	-0.0380	-3.7319	-0.0350	-3.4348
dist_meadow_log	0.0067	0.6760	0.0082	0.8205	0.0081	0.8090	0.0081	0.8108
dist_np_log	-0.0646	-6.2533	-0.0515	-5.0219	-0.0554	-5.3916	-0.0516	-5.0328
dist_ropeway_log_ max_elevat	-0.0514	-5.0078	-0.0397	-3.8932	-0.0565	-5.4894	-0.0541	-5.2696
dist_stadium_log	0.0089	0.8931	0.0048	0.4851	0.0083	0.8303	0.0073	0.7349
dist_synagogue_log	-0.0627	-6.0748	-0.0525	-5.1141	-0.0570	-5.5364	-0.0586	-5.6903
dist_tailings_ dump_log	0.0163	1.6414	0.0144	1.4456	0.0123	1.2383	0.0147	1.4802
dist_water_area_ log	-0.0151	-1.4994	-0.0134	-1.3292	-0.0091	-0.9028	-0.0106	-1.0571
dist_water_flow_ log	0.0201	2.0317	0.0176	1.7705	0.0171	1.7296	0.0183	1.8435
elevator	0.0114	1.1447	0.0038	0.3852	0.0021	0.2078	0.0048	0.4795
equipped	0.0061	0.6133	0.0019	0.1950	0.0020	0.1976	0.0032	0.3186
floornumber	0.0038	0.3759	-0.0033	-0.3274	-0.0010	-0.1043	-0.0031	-0.3122
floornumber_elevator	-0.0087	-0.8688	-0.0068	-0.6814	-0.0070	-0.6957	-0.0076	-0.7584
garage	-0.0075	-0.7490	-0.0047	-0.4717	-0.0066	-0.6531	-0.0062	-0.6169
garden	0.0025	0.2462	0.0046	0.4615	-0.0004	-0.0421	0.0008	0.0767
loggia	0.0094	0.9443	0.0020	0.2019	0.0058	0.5807	0.0025	0.2494
own_private	0.0872	9.1163	0.0817	8.5141	0.0731	7.5823	0.0710	7.3597
parking_space	0.0051	0.5158	-0.0037	-0.3706	0.0012	0.1231	0.0011	0.1051
pop_tot_log_dist_ city_cen_log	0.0479	4.9048	0.0577	5.9365	0.0339	3.4434	0.0284	2.8796
population_total_ log	-0.1904	-17.3399	-0.2222	-19.9207	-0.1347	-12.6057	-0.1245	-11.7081

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feature	no embeddings		TF-IDF		Word2Vec		BERT	
	coeff	$\Delta\%$	coeff	$\Delta\%$	coeff	$\Delta\%$	coeff	$\Delta\%$
rooms	0.0505	5.1826	0.0565	5.8146	0.0626	6.4568	0.0564	5.7969
sep_kitchen	-0.0818	-7.8528	-0.0519	-5.0619	-0.0551	-5.3620	-0.0540	-5.2611
terrace	0.0246	2.4890	0.0147	1.4789	0.0170	1.7151	0.0128	1.2916
usable_area_log	-0.1055	-10.0107	-0.1385	-12.9304	-0.1359	-12.7101	-0.1346	-12.5909
municipality code:								
500496	0.0023	0.2297	0.0092	0.9282	0.0011	0.1129	0.0012	0.1161
532053	0.0399	4.0743	0.0346	3.5170	0.0350	3.5603	0.0327	3.3197
544256	0.0179	1.8111	0.0214	2.1620	0.0191	1.9323	0.0195	1.9718
554782	0.3455	41.2672	0.3226	38.0757	0.2912	33.7986	0.2835	32.7762
554791	0.0232	2.3453	0.0294	2.9882	0.0233	2.3609	0.0251	2.5383
554804	-0.0730	-7.0368	-0.0602	-5.8396	-0.0627	-6.0795	-0.0606	-5.8809
554821	-0.0110	-1.0943	0.0024	0.2360	-0.0165	-1.6378	-0.0144	-1.4324
554961	-0.0227	-2.2408	-0.0191	-1.8896	-0.0249	-2.4620	-0.0226	-2.2326
555134	0.0026	0.2590	0.0088	0.8827	0.0044	0.4410	0.0038	0.3840
563889	-0.0020	-0.2035	0.0016	0.1646	-0.0006	-0.0626	-0.0009	-0.0853
567027	-0.0985	-9.3814	-0.0856	-8.2032	-0.0919	-8.7827	-0.0892	-8.5358
567442	-0.0618	-5.9885	-0.0541	-5.2698	-0.0554	-5.3853	-0.0562	-5.4679
569810	0.0034	0.3405	0.0104	1.0410	0.0043	0.4261	0.0038	0.3821
582786	0.0716	7.4214	0.0726	7.5300	0.0597	6.1556	0.0572	5.8855
date:								
2014 Q2	-0.0023	-0.2274	-0.0024	-0.2376	-0.0016	-0.1608	-0.0008	-0.0835
2014 Q3	0.0016	0.1620	-0.0003	-0.0270	0.0035	0.3466	0.0032	0.3204
2014 Q4	-0.0000	-0.0044	-0.0014	-0.1442	0.0025	0.2490	0.0030	0.2975
2015 Q1	0.0048	0.4847	0.0033	0.3295	0.0074	0.7416	0.0080	0.8071
2015 Q2	0.0093	0.9391	0.0072	0.7261	0.0106	1.0668	0.0107	1.0754
2015 Q3	0.0092	0.9217	0.0077	0.7711	0.0105	1.0512	0.0105	1.0603
2015 Q4	0.0088	0.8840	0.0079	0.7975	0.0102	1.0238	0.0113	1.1385
2016 Q1	0.0153	1.5420	0.0145	1.4591	0.0170	1.7170	0.0179	1.8056
2016 Q2	0.0239	2.4222	0.0220	2.2277	0.0251	2.5448	0.0256	2.5920
2016 Q3	0.0268	2.7187	0.0250	2.5267	0.0280	2.8418	0.0285	2.8879
2016 Q4	0.0324	3.2923	0.0304	3.0913	0.0334	3.3918	0.0336	3.4191
2017 Q1	0.0397	4.0458	0.0377	3.8375	0.0399	4.0694	0.0409	4.1763
2017 Q2	0.0462	4.7301	0.0441	4.5082	0.0467	4.7757	0.0474	4.8563
2017 Q3	0.0485	4.9739	0.0460	4.7093	0.0484	4.9559	0.0489	5.0151
2017 Q4	0.0508	5.2093	0.0485	4.9726	0.0510	5.2287	0.0515	5.2850
2018 Q1	0.0499	5.1202	0.0480	4.9201	0.0502	5.1522	0.0508	5.2115
2018 Q2	0.0519	5.3283	0.0495	5.0698	0.0515	5.2869	0.0519	5.3297
2018 Q3	0.0556	5.7194	0.0534	5.4861	0.0556	5.7125	0.0559	5.7469
2018 Q4	0.0578	5.9460	0.0554	5.6913	0.0577	5.9360	0.0579	5.9610
2019 Q1	0.0610	6.2936	0.0586	6.0333	0.0608	6.2738	0.0613	6.3218
2019 Q2	0.0898	9.3979	0.0863	9.0114	0.0892	9.3304	0.0896	9.3717
2019 Q3	0.0743	7.7157	0.0716	7.4174	0.0741	7.6883	0.0749	7.7728
2019 Q4	0.0787	8.1856	0.0758	7.8765	0.0785	8.1638	0.0795	8.2750
2020 Q1	0.0807	8.4087	0.0777	8.0813	0.0806	8.3889	0.0813	8.4657
2020 Q2	0.0871	9.1052	0.0828	8.6377	0.0852	8.8984	0.0856	8.9352
2020 Q3	0.0982	10.3145	0.0936	9.8084	0.0956	10.0274	0.0956	10.0331
2020 Q4	0.1016	10.6937	0.0974	10.2327	0.0995	10.4597	0.0997	10.4812
2021 Q1	0.1059	11.1666	0.1016	10.6895	0.1036	10.9109	0.1044	10.9993
2021 Q2	0.1116	11.8077	0.1073	11.3268	0.1086	11.4693	0.1095	11.5762
2021 Q3	0.1137	12.0452	0.1096	11.5827	0.1114	11.7787	0.1115	11.7986
2021 Q4	0.1226	13.0465	0.1186	12.5885	0.1200	12.7445	0.1209	12.8514
2022 Q1	0.1308	13.9753	0.1268	13.5179	0.1277	13.6156	0.1288	13.7413
2022 Q2	0.1377	14.7650	0.1338	14.3189	0.1330	14.2250	0.1342	14.3602
2022 Q3	0.1402	15.0448	0.1363	14.6076	0.1344	14.3846	0.1352	14.4716
2022 Q4	0.1576	17.0688	0.1535	16.5918	0.1516	16.3675	0.1526	16.4837
2023 Q1	0.1569	16.9905	0.1530	16.5359	0.1508	16.2775	0.1517	16.3760

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feature	no embeddings		TF-IDF		Word2Vec		BERT	
	coeff	$\Delta\%$	coeff	$\Delta\%$	coeff	$\Delta\%$	coeff	$\Delta\%$
2023 Q2	0.1474	15.8861	0.1436	15.4434	0.1414	15.1923	0.1430	15.3704
2023 Q3	0.1410	15.1378	0.1370	14.6777	0.1345	14.3918	0.1357	14.5334
2023 Q4	0.1476	15.9086	0.1429	15.3635	0.1402	15.0489	0.1416	15.2120
2024 Q1	0.1482	15.9749	0.1435	15.4258	0.1411	15.1533	0.1425	15.3195
2024 Q2	0.1550	16.7614	0.1491	16.0812	0.1469	15.8231	0.1484	15.9969
2024 Q3	0.1452	15.6258	0.1400	15.0267	0.1383	14.8264	0.1386	14.8671
2024 Q4	0.1151	12.1964	0.1103	11.6604	0.1090	11.5185	0.1080	11.4082
weekday:								
Monday	-0.0012	-0.1220	-0.0006	-0.0557	-0.0011	-0.1120	-0.0007	-0.0731
Tuesday	-0.0027	-0.2656	-0.0017	-0.1706	-0.0018	-0.1846	-0.0016	-0.1629
Wednesday	-0.0005	-0.0479	-0.0002	-0.0220	-0.0002	-0.0169	-0.0005	-0.0492
Thursday	-0.0010	-0.0965	-0.0012	-0.1174	-0.0009	-0.0942	-0.0013	-0.1280
Friday	0.0015	0.1497	0.0013	0.1283	0.0013	0.1340	0.0015	0.1547
Saturday	-0.0017	-0.1666	-0.0015	-0.1474	-0.0018	-0.1839	-0.0014	-0.1428
created_after__ invasion	-0.0081	-0.8045	-0.0119	-1.1786	-0.0005	-0.0485	0.0008	0.0837
created_after__ pandemic	0.0045	0.4469	0.0029	0.2902	0.0088	0.8811	0.0094	0.9443

Note: Columns show coefficients for each scenario; $\Delta\%$ is the implied percentage change.

Appendix E

Results of ML Methods

E.1 Random Forest

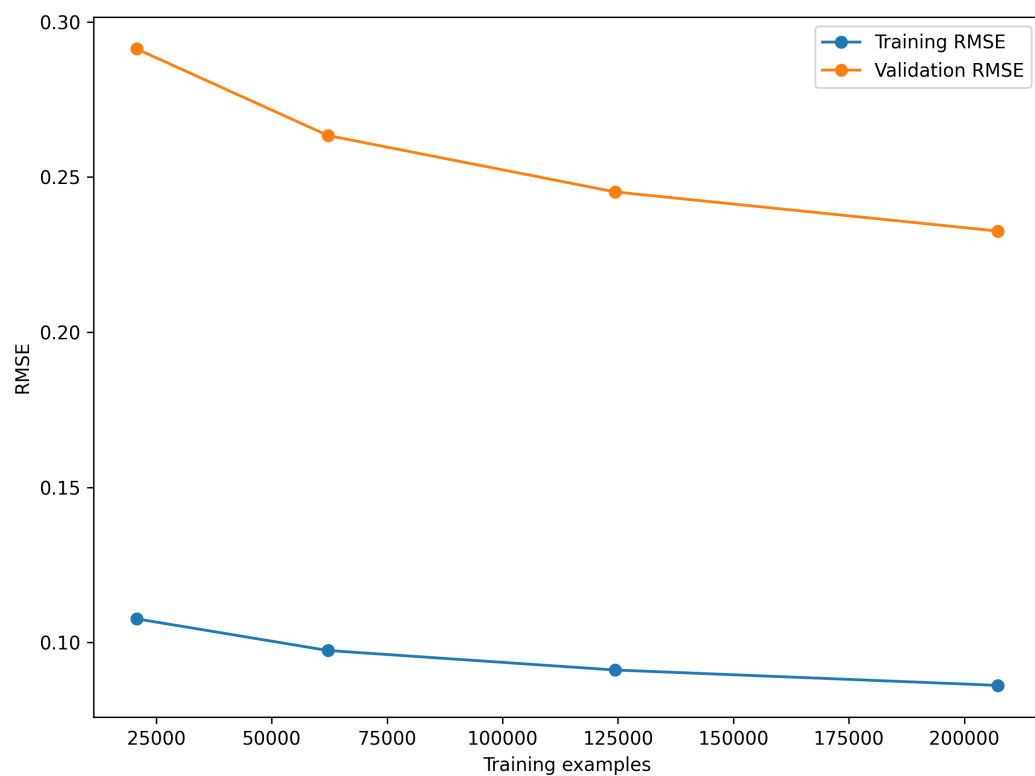


Figure E.1: Random Forest learning curve (baseline scenario)

Note: Computed on a random 20% subset of the data. Learning curves of other scenarios follow similar patterns.

Table E.1: RF variable importance comparison

feature	no embeddings		TF-IDF		Word2Vec		BERT	
	perm	gini	perm	gini	perm	gini	perm	gini
advertisement_ length_log	0.0100	0.0110	0.0067	0.0050	0.0076	0.0058	0.0082	0.0061
balcony	0.0025	0.0040	0.0001	0.0002	0.0001	0.0002	0.0002	0.0003
barrier_free	0.0014	0.0011	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
bldg_mixed	0.0010	0.0018	0.0003	0.0003	0.0001	0.0002	0.0002	0.0003
bldg_prefabricated	0.0135	0.0159	0.0060	0.0053	0.0145	0.0107	0.0139	0.0101
bldg_skeleton	0.0001	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
cond_ before refurbishment	0.0008	0.0010	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004
cond_good	0.0102	0.0095	0.0035	0.0031	0.0048	0.0038	0.0047	0.0035
cond_new development	0.0070	0.0102	0.0015	0.0027	0.0060	0.0069	0.0057	0.0061
cond_project	0.0001	0.0004	0.0000	0.0000	0.0000	0.0001	0.0000	0.0001
cond_ under construction	0.0013	0.0018	0.0001	0.0002	0.0003	0.0005	0.0003	0.0006
cond_very good	0.0020	0.0014	0.0001	0.0001	0.0002	0.0002	0.0002	0.0002
dist_airport_log	0.0624	0.0599	0.0931	0.0565	0.1009	0.0608	0.1018	0.0596
dist_castle_log	0.0287	0.0222	0.0270	0.0150	0.0283	0.0156	0.0289	0.0157
dist_chateau_log	0.0165	0.0138	0.0112	0.0079	0.0117	0.0081	0.0129	0.0087
dist_church_log	0.0062	0.0078	0.0026	0.0029	0.0026	0.0030	0.0025	0.0028
dist_city_center_ log	0.0126	0.0229	0.0021	0.0046	0.0022	0.0038	0.0024	0.0038
dist_closest_ county_center_log	0.0147	0.0118	0.0083	0.0062	0.0083	0.0062	0.0089	0.0064
dist_closest_ regional_center_log	0.0345	0.0392	0.0188	0.0178	0.0191	0.0175	0.0216	0.0184
dist_forest_log	0.0080	0.0073	0.0024	0.0025	0.0028	0.0028	0.0028	0.0029
dist_highway_log	0.0266	0.0244	0.0182	0.0147	0.0170	0.0136	0.0177	0.0132
dist_hospital_log	0.0152	0.0115	0.0078	0.0055	0.0075	0.0055	0.0084	0.0057
dist_meadow_log	0.0088	0.0109	0.0030	0.0030	0.0030	0.0029	0.0029	0.0029
dist_np_log	0.0311	0.0302	0.0272	0.0182	0.0281	0.0185	0.0306	0.0188
dist_ropeway_log_ max_elevat	0.0655	0.0781	0.0565	0.0766	0.0638	0.0778	0.0647	0.0792
dist_stadium_log	0.0100	0.0093	0.0040	0.0036	0.0047	0.0040	0.0047	0.0039
dist_synagogue_log	0.0558	0.0674	0.0690	0.0635	0.0711	0.0665	0.0663	0.0661
dist_tailings_ dump_log	0.0140	0.0122	0.0073	0.0065	0.0071	0.0064	0.0069	0.0061
dist_water_area_ log	0.0060	0.0074	0.0020	0.0027	0.0020	0.0028	0.0020	0.0027
dist_water_flow_ log	0.0058	0.0067	0.0011	0.0013	0.0013	0.0015	0.0014	0.0016
elevator	0.0008	0.0008	0.0000	0.0001	0.0001	0.0001	0.0001	0.0001
equipped	0.0011	0.0017	0.0001	0.0001	0.0001	0.0001	0.0002	0.0002
floornumber	0.0028	0.0031	0.0004	0.0004	0.0003	0.0004	0.0004	0.0004
floornumber_elevator	0.0012	0.0015	0.0001	0.0002	0.0001	0.0002	0.0001	0.0002
garage	0.0010	0.0013	0.0000	0.0001	0.0000	0.0001	0.0000	0.0001
garden	0.0002	0.0004	0.0000	0.0000	0.0000	0.0001	0.0000	0.0001
loggia	0.0067	0.0048	0.0011	0.0006	0.0020	0.0010	0.0016	0.0008
own_private	0.0252	0.0229	0.0235	0.0163	0.0281	0.0205	0.0289	0.0217
parking_space	0.0005	0.0008	0.0000	0.0001	0.0001	0.0001	0.0001	0.0001
pop_tot_log_dist_ city_cen_log	0.0143	0.0245	0.0038	0.0118	0.0038	0.0115	0.0041	0.0121
population_total_ log	0.0940	0.0951	0.1406	0.1500	0.1508	0.1521	0.1454	0.1516

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feature	no embeddings		TF-IDF		Word2Vec		BERT	
	perm	gini	perm	gini	perm	gini	perm	gini
rooms	0.0039	0.0034	0.0016	0.0013	0.0014	0.0012	0.0016	0.0012
sep_kitchen	0.0378	0.0471	0.0353	0.0437	0.0523	0.0609	0.0525	0.0580
terrace	0.0014	0.0011	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
usable_area_log	0.0170	0.0112	0.0195	0.0083	0.0178	0.0076	0.0184	0.0078
municipality code:								
500496	0.0001	0.0004	0.0000	0.0001	0.0000	0.0001	0.0000	0.0001
532053	0.0001	0.0003	0.0000	0.0001	0.0000	0.0000	0.0000	0.0000
544256	0.0001	0.0005	0.0001	0.0004	0.0001	0.0004	0.0001	0.0003
554782	0.0386	0.0563	0.0433	0.0848	0.0469	0.0867	0.0446	0.0854
554791	0.0001	0.0006	0.0000	0.0002	0.0001	0.0002	0.0001	0.0003
554804	0.0009	0.0017	0.0004	0.0010	0.0004	0.0011	0.0003	0.0008
554821	0.0007	0.0026	0.0002	0.0004	0.0001	0.0003	0.0002	0.0004
554961	0.0002	0.0005	0.0001	0.0001	0.0000	0.0001	0.0001	0.0001
555134	0.0002	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
563889	0.0000	0.0002	0.0000	0.0001	0.0000	0.0001	0.0000	0.0001
567027	0.0022	0.0055	0.0012	0.0028	0.0015	0.0032	0.0017	0.0030
567442	0.0010	0.0020	0.0001	0.0004	0.0001	0.0004	0.0001	0.0004
569810	0.0001	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
582786	0.0020	0.0043	0.0012	0.0051	0.0014	0.0052	0.0012	0.0052
date:								
2014 Q2	0.0014	0.0019	0.0009	0.0007	0.0010	0.0007	0.0011	0.0008
2014 Q3	0.0027	0.0037	0.0026	0.0016	0.0028	0.0016	0.0031	0.0018
2014 Q4	0.0031	0.0027	0.0025	0.0014	0.0026	0.0015	0.0029	0.0016
2015 Q1	0.0034	0.0038	0.0035	0.0019	0.0038	0.0020	0.0043	0.0023
2015 Q2	0.0022	0.0023	0.0020	0.0012	0.0023	0.0012	0.0025	0.0014
2015 Q3	0.0004	0.0007	0.0005	0.0004	0.0005	0.0004	0.0006	0.0004
2015 Q4	0.0008	0.0010	0.0008	0.0005	0.0008	0.0005	0.0009	0.0006
2016 Q1	0.0006	0.0008	0.0006	0.0004	0.0006	0.0004	0.0007	0.0004
2016 Q2	0.0004	0.0006	0.0003	0.0002	0.0004	0.0003	0.0004	0.0003
2016 Q3	0.0001	0.0004	0.0001	0.0001	0.0002	0.0001	0.0002	0.0002
2016 Q4	0.0002	0.0003	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
2017 Q1	0.0001	0.0003	0.0000	0.0001	0.0000	0.0001	0.0001	0.0001
2017 Q2	0.0002	0.0003	0.0000	0.0001	0.0000	0.0001	0.0000	0.0001
2017 Q3	0.0001	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2017 Q4	0.0002	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001
2018 Q1	0.0003	0.0003	0.0000	0.0001	0.0001	0.0001	0.0001	0.0001
2018 Q2	0.0002	0.0003	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
2018 Q3	0.0005	0.0004	0.0001	0.0001	0.0001	0.0001	0.0001	0.0002
2018 Q4	0.0003	0.0004	0.0001	0.0001	0.0002	0.0002	0.0002	0.0002
2019 Q1	0.0003	0.0005	0.0002	0.0002	0.0003	0.0003	0.0003	0.0003
2019 Q2	0.0010	0.0013	0.0012	0.0011	0.0013	0.0011	0.0014	0.0012
2019 Q3	0.0014	0.0011	0.0010	0.0010	0.0012	0.0010	0.0012	0.0011
2019 Q4	0.0014	0.0014	0.0016	0.0013	0.0017	0.0014	0.0019	0.0015
2020 Q1	0.0028	0.0016	0.0020	0.0016	0.0022	0.0017	0.0024	0.0018
2020 Q2	0.0006	0.0011	0.0009	0.0008	0.0011	0.0009	0.0010	0.0009
2020 Q3	0.0006	0.0010	0.0007	0.0006	0.0007	0.0006	0.0007	0.0006
2020 Q4	0.0003	0.0008	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004
2021 Q1	0.0001	0.0007	0.0001	0.0002	0.0002	0.0002	0.0001	0.0002
2021 Q2	0.0002	0.0009	0.0001	0.0001	0.0001	0.0001	0.0001	0.0002
2021 Q3	0.0006	0.0011	0.0001	0.0002	0.0001	0.0002	0.0001	0.0003
2021 Q4	0.0005	0.0017	0.0004	0.0006	0.0004	0.0006	0.0004	0.0006
2022 Q1	0.0005	0.0020	0.0006	0.0008	0.0007	0.0008	0.0007	0.0009
2022 Q2	0.0005	0.0008	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
2022 Q3	0.0002	0.0006	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2022 Q4	0.0003	0.0009	0.0000	0.0001	0.0000	0.0000	0.0000	0.0000
2023 Q1	0.0002	0.0007	0.0000	0.0001	0.0000	0.0000	0.0000	0.0000

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feature	no embeddings		TF-IDF		Word2Vec		BERT	
	perm	gini	perm	gini	perm	gini	perm	gini
2023 Q2	0.0006	0.0004	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2023 Q3	0.0002	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2023 Q4	0.0000	0.0004	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2024 Q1	0.0001	0.0005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2024 Q2	0.0002	0.0008	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2024 Q3	0.0001	0.0006	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2024 Q4	0.0000	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
weekday:								
Monday	0.0005	0.0008	0.0000	0.0001	0.0000	0.0001	0.0000	0.0001
Tuesday	0.0004	0.0008	0.0000	0.0001	0.0000	0.0001	0.0000	0.0001
Wednesday	0.0006	0.0008	0.0000	0.0001	0.0000	0.0001	0.0000	0.0001
Thursday	0.0003	0.0007	0.0000	0.0001	0.0000	0.0001	0.0000	0.0001
Friday	0.0002	0.0004	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Saturday	0.0003	0.0006	0.0000	0.0001	0.0000	0.0001	0.0000	0.0000
created_after__ invasion	0.0485	0.0510	0.0475	0.0498	0.0481	0.0523	0.0583	0.0602
created_after__ pandemic	0.1857	0.1066	0.2649	0.1244	0.2685	0.1310	0.2586	0.1228

Note: perm is permutation importance; gini is mean decrease in impurity.

E.2 XGBoost

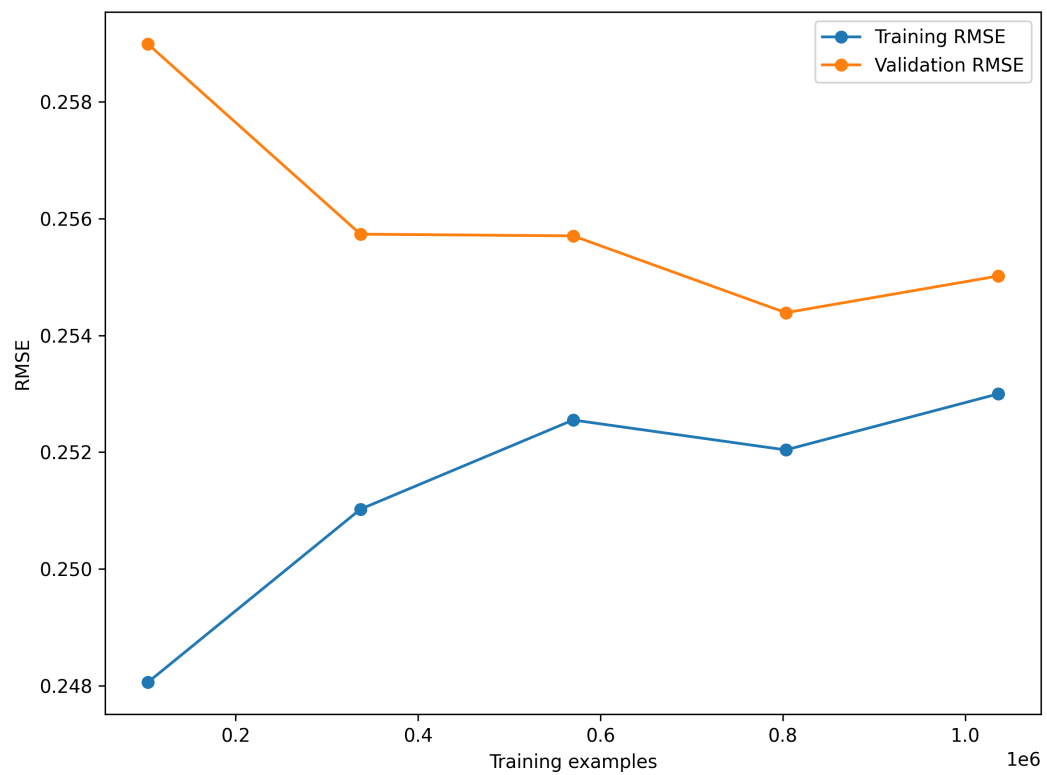


Figure E.2: XGBoost learning curve (baseline scenario)

Note: Computed on a random 20% subset of the data. Learning curves of other scenarios follow similar patterns.

Table E.2: XGB variable importance comparison

feature	no embeddings		TF-IDF		Word2Vec		BERT	
	perm	gain	perm	gain	perm	gain	perm	gain
advertisement_								
length_log	0.0046	0.0018	0.0037	0.0022	0.0041	0.0011	0.0041	0.0012
balcony	0.0017	0.0024	0.0003	0.0008	0.0010	0.0012	0.0009	0.0014
barrier_free	0.0012	0.0015	0.0005	0.0026	0.0007	0.0016	0.0006	0.0016
bldg_mixed	0.0014	0.0009	0.0003	0.0011	0.0002	0.0008	0.0001	0.0007
bldg_prefabricated	0.0053	0.0086	0.0019	0.0060	0.0059	0.0071	0.0061	0.0075
bldg_skeleton	0.0000	0.0002	0.0000	0.0002	0.0000	0.0002	0.0000	0.0002
cond_								
before refurbishment	0.0045	0.0030	0.0021	0.0030	0.0031	0.0048	0.0028	0.0054
cond_good	0.0085	0.0115	0.0033	0.0287	0.0045	0.0086	0.0046	0.0090
cond_new development	0.0055	0.0042	0.0012	0.0009	0.0047	0.0086	0.0037	0.0085
cond_project	0.0002	0.0005	0.0000	0.0001	0.0001	0.0004	0.0001	0.0004
cond_								
under construction	0.0004	0.0008	0.0001	0.0005	0.0004	0.0012	0.0003	0.0009
cond_very good	0.0018	0.0017	0.0004	0.0005	0.0008	0.0013	0.0007	0.0010
dist_airport_log	0.0212	0.0125	0.0219	0.0177	0.0187	0.0047	0.0184	0.0049
dist_castle_log	0.0068	0.0018	0.0052	0.0032	0.0087	0.0024	0.0087	0.0027
dist_chateau_log	0.0016	0.0008	0.0009	0.0008	0.0021	0.0009	0.0020	0.0008
dist_church_log	0.0004	0.0008	0.0005	0.0009	0.0009	0.0003	0.0008	0.0004
dist_city_center_								
log	0.0006	0.0006	0.0005	0.0008	0.0010	0.0003	0.0010	0.0003
dist_city_center_								
log_2	0.0003	0.0008	0.0002	0.0006	0.0004	0.0011	0.0004	0.0012
dist_closest_								
county_center_log	0.0029	0.0011	0.0020	0.0013	0.0015	0.0006	0.0016	0.0006
dist_closest_								
regional_center_log	0.0191	0.0044	0.0141	0.0056	0.0137	0.0028	0.0140	0.0032
dist_forest_log	0.0003	0.0006	0.0002	0.0013	0.0007	0.0004	0.0007	0.0005
dist_highway_log	0.0071	0.0033	0.0049	0.0036	0.0073	0.0026	0.0071	0.0027
dist_hospital_log	0.0063	0.0024	0.0055	0.0030	0.0056	0.0012	0.0049	0.0012
dist_meadow_log	0.0025	0.0039	0.0026	0.0043	0.0016	0.0007	0.0015	0.0008
dist_np_log	0.0219	0.0041	0.0231	0.0052	0.0270	0.0044	0.0259	0.0044
dist_ropeway_log_								
max_elevat	0.0215	0.0205	0.0186	0.0253	0.0196	0.0061	0.0195	0.0068
dist_stadium_log	0.0007	0.0007	0.0005	0.0010	0.0014	0.0006	0.0014	0.0007
dist_synagogue_log	0.0161	0.0190	0.0115	0.0244	0.0125	0.0084	0.0124	0.0093
dist_tailings_								
dump_log	0.0021	0.0015	0.0010	0.0011	0.0021	0.0010	0.0020	0.0011
dist_water_area_								
log	0.0009	0.0009	0.0005	0.0010	0.0007	0.0003	0.0007	0.0004
dist_water_flow_								
log	0.0003	0.0003	0.0001	0.0002	0.0007	0.0002	0.0006	0.0003
elevator	0.0001	0.0003	0.0000	0.0003	0.0001	0.0002	0.0001	0.0002
equipped	0.0001	0.0042	0.0000	0.0004	0.0000	0.0012	0.0000	0.0011
floornumber	0.0002	0.0004	0.0001	0.0003	0.0006	0.0002	0.0006	0.0002
floornumber_elevator	0.0000	0.0002	0.0000	0.0002	0.0001	0.0001	0.0001	0.0002
garage	0.0000	0.0003	0.0000	0.0000	0.0000	0.0002	0.0000	0.0003
garden	0.0002	0.0003	0.0000	0.0002	0.0001	0.0003	0.0001	0.0003
loggia	0.0008	0.0072	0.0002	0.0049	0.0006	0.0038	0.0005	0.0044
own_private	0.0164	0.0183	0.0133	0.0215	0.0133	0.0115	0.0136	0.0121
parking_space	0.0009	0.0008	0.0002	0.0006	0.0009	0.0008	0.0008	0.0009
pop_tot_log_dist_								
city_cen_log	0.0005	0.0010	0.0003	0.0021	0.0011	0.0013	0.0010	0.0014

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feature	no embeddings		TF-IDF		Word2Vec		BERT	
	perm	gain	perm	gain	perm	gain	perm	gain
pop_tot_log_dist_ city_cen_log_2	0.0001	0.0006	0.0001	0.0007	0.0003	0.0003	0.0003	0.0004
population_total_ log	0.1201	0.0551	0.1224	0.0683	0.1670	0.0501	0.1654	0.0543
rooms	0.0135	0.0008	0.0114	0.0010	0.0203	0.0008	0.0204	0.0008
sep_kitchen	0.0139	0.0252	0.0071	0.0201	0.0089	0.0265	0.0083	0.0250
terrace	0.0008	0.0028	0.0002	0.0015	0.0006	0.0010	0.0005	0.0011
usable_area_log	0.0275	0.0015	0.0333	0.0022	0.0518	0.0011	0.0526	0.0012
usable_area_log_2	0.0058	0.0018	0.0045	0.0017	0.0053	0.0010	0.0054	0.0011
region code:								
1 08	0.0004	0.0007	0.0003	0.0011	0.0004	0.0035	0.0004	0.0030
116	0.0032	0.0040	0.0023	0.0021	0.0021	0.0045	0.0022	0.0047
124	0.0005	0.0008	0.0003	0.0006	0.0004	0.0005	0.0004	0.0006
132	0.0009	0.0254	0.0009	0.0365	0.0013	0.0193	0.0013	0.0189
141	0.0016	0.0036	0.0021	0.0041	0.0013	0.0051	0.0012	0.0047
27	0.0100	0.0065	0.0069	0.0047	0.0061	0.0116	0.0062	0.0115
35	0.0001	0.0007	0.0001	0.0013	0.0001	0.0003	0.0001	0.0004
43	0.0000	0.0002	0.0000	0.0006	0.0001	0.0002	0.0001	0.0003
51	0.0110	0.0033	0.0086	0.0030	0.0085	0.0036	0.0089	0.0038
60	0.0570	0.0325	0.0513	0.0285	0.0564	0.0364	0.0565	0.0396
78	0.0005	0.0003	0.0005	0.0004	0.0009	0.0007	0.0009	0.0007
86	0.0001	0.0006	0.0000	0.0004	0.0002	0.0005	0.0001	0.0004
94	0.0006	0.0011	0.0004	0.0010	0.0005	0.0007	0.0005	0.0007
municipality code:								
500496	0.0000	0.0002	0.0000	0.0002	0.0000	0.0108	0.0000	0.0180
532053	0.0000	0.0002	0	0	0.0000	0.0002	0.0000	0.0002
544256	0.0005	0.0032	0.0002	0.0032	0.0000	0.0015	0.0000	0.0018
554782	0.0090	0.3535	0.0054	0.2407	0.0137	0.3185	0.0130	0.2956
554791	0	0	0.0000	0.0001	0.0000	0.0002	0.0000	0.0001
554804	0.0005	0.0015	0.0003	0.0017	0.0000	0.0036	0.0000	0.0034
554821	0.0000	0.0003	0.0000	0.0080	0.0000	0.0012	0.0000	0.0022
554961	0.0013	0.0004	0.0011	0.0005	0.0001	0.0004	0.0001	0.0004
555134	0	0	0	0	0.0000	0.0001	0.0000	0.0001
563889	0.0000	0.0004	0.0000	0.0005	0.0000	0.0002	0.0000	0.0001
567027	0.0009	0.0018	0.0009	0.0020	0.0006	0.0047	0.0005	0.0043
567442	0.0002	0.0007	0.0002	0.0007	0.0003	0.0007	0.0005	0.0010
569810	0	0	0	0	0.0000	0.0001	0.0000	0.0001
582786	0.0015	0.0005	0.0015	0.0010	0.0001	0.0159	0.0002	0.0172
municipality district code:								
500089	0.0003	0.0005	0.0001	0.0004	0.0001	0.0003	0.0001	0.0003
500097	0	0	0	0	0.0000	0.0001	0.0000	0.0001
500119	0.0000	0.0012	0	0	0.0001	0.0003	0.0001	0.0003
500143	0.0000	0.0003	0.0000	0.0005	0.0000	0.0002	0.0000	0.0005
500178	0.0003	0.0005	0.0003	0.0009	0.0005	0.0008	0.0005	0.0008
500186	0.0000	0.0001	0	0	0.0000	0.0002	0.0000	0.0011
500208	0.0001	0.0006	0.0000	0.0005	0.0000	0.0002	0.0000	0.0003
500216	0.0001	0.0005	0.0001	0.0004	0.0000	0.0001	0.0000	0.0002
500224	0	0	0.0000	0.0005	0.0000	0.0003	0.0000	0.0004
539694	0.0000	0.0002	0.0000	0.0003	0.0000	0.0001	0.0000	0.0002
545911	0.0000	0.0002	0	0	0.0000	0.0003	0.0000	0.0003
546135	0.0012	0.0012	0.0013	0.0011	0.0005	0.0015	0.0005	0.0013
546224	0.0000	0.0002	0	0	0.0000	0.0005	0.0001	0.0006
547034	0	0	0	0	0.0000	0.0001	0.0000	0.0001
547107	0.0000	0.0002	0.0000	0.0009	0.0000	0.0001	0.0000	0.0006
550973	0	0	0.0000	0.0001	0.0000	0.0001	0.0000	0.0002

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feature	no embeddings		TF-IDF		Word2Vec		BERT	
	perm	gain	perm	gain	perm	gain	perm	gain
date:								
2014 Q2	0.0038	0.0081	0.0035	0.0080	0.0033	0.0067	0.0032	0.0069
2014 Q3	0.0054	0.0103	0.0054	0.0101	0.0049	0.0071	0.0048	0.0085
2014 Q4	0.0053	0.0094	0.0050	0.0085	0.0045	0.0113	0.0045	0.0105
2015 Q1	0.0064	0.0109	0.0060	0.0096	0.0051	0.0122	0.0052	0.0122
2015 Q2	0.0043	0.0080	0.0042	0.0075	0.0037	0.0102	0.0037	0.0095
2015 Q3	0.0018	0.0039	0.0018	0.0039	0.0016	0.0067	0.0016	0.0071
2015 Q4	0.0023	0.0051	0.0021	0.0051	0.0019	0.0097	0.0019	0.0096
2016 Q1	0.0017	0.0043	0.0016	0.0042	0.0014	0.0087	0.0014	0.0094
2016 Q2	0.0009	0.0031	0.0010	0.0027	0.0007	0.0058	0.0007	0.0059
2016 Q3	0.0004	0.0016	0.0004	0.0015	0.0003	0.0028	0.0004	0.0038
2016 Q4	0.0002	0.0007	0.0002	0.0008	0.0001	0.0010	0.0001	0.0012
2017 Q1	0.0000	0.0003	0.0000	0.0003	0.0001	0.0003	0.0001	0.0003
2017 Q2	0.0001	0.0003	0.0001	0.0003	0.0003	0.0005	0.0003	0.0005
2017 Q3	0.0003	0.0006	0.0002	0.0006	0.0004	0.0009	0.0005	0.0011
2017 Q4	0.0005	0.0008	0.0004	0.0008	0.0007	0.0011	0.0007	0.0015
2018 Q1	0.0006	0.0010	0.0006	0.0010	0.0009	0.0016	0.0009	0.0018
2018 Q2	0.0009	0.0015	0.0008	0.0015	0.0012	0.0025	0.0012	0.0029
2018 Q3	0.0012	0.0018	0.0011	0.0019	0.0015	0.0033	0.0016	0.0036
2018 Q4	0.0015	0.0022	0.0014	0.0022	0.0017	0.0040	0.0018	0.0047
2019 Q1	0.0020	0.0027	0.0018	0.0027	0.0023	0.0045	0.0023	0.0048
2019 Q2	0.0045	0.0065	0.0040	0.0054	0.0043	0.0077	0.0043	0.0081
2019 Q3	0.0039	0.0044	0.0035	0.0044	0.0040	0.0078	0.0040	0.0087
2019 Q4	0.0048	0.0053	0.0043	0.0049	0.0049	0.0075	0.0049	0.0084
2020 Q1	0.0048	0.0047	0.0044	0.0048	0.0055	0.0084	0.0055	0.0092
2020 Q2	0.0013	0.0021	0.0009	0.0025	0.0032	0.0029	0.0030	0.0029
2020 Q3	0.0006	0.0020	0.0004	0.0017	0.0011	0.0025	0.0010	0.0021
2020 Q4	0.0002	0.0022	0.0002	0.0015	0.0007	0.0022	0.0008	0.0024
2021 Q1	0.0000	0.0004	0.0000	0.0007	0.0002	0.0013	0.0002	0.0009
2021 Q2	0.0009	0.0004	0.0009	0.0004	0.0012	0.0004	0.0009	0.0004
2021 Q3	0.0021	0.0008	0.0014	0.0009	0.0019	0.0016	0.0015	0.0018
2021 Q4	0.0026	0.0014	0.0022	0.0017	0.0023	0.0029	0.0020	0.0032
2022 Q1	0.0030	0.0018	0.0028	0.0021	0.0030	0.0043	0.0025	0.0047
2022 Q2	0.0019	0.0012	0.0018	0.0013	0.0020	0.0023	0.0020	0.0026
2022 Q3	0.0012	0.0009	0.0012	0.0010	0.0016	0.0014	0.0016	0.0015
2022 Q4	0.0008	0.0004	0.0006	0.0005	0.0009	0.0006	0.0009	0.0007
2023 Q1	0.0001	0.0004	0.0000	0.0005	0.0002	0.0003	0.0001	0.0002
2023 Q2	0.0000	0.0001	0	0	0.0001	0.0005	0.0000	0.0005
2023 Q3	0.0000	0.0001	0.0000	0.0002	0.0001	0.0003	0.0001	0.0003
2023 Q4	0	0	0	0	0.0001	0.0002	0.0000	0.0002
2024 Q1	0.0000	0.0001	0	0	0.0001	0.0001	0.0002	0.0001
2024 Q2	0.0003	0.0004	0.0002	0.0004	0.0003	0.0005	0.0003	0.0006
2024 Q3	0.0005	0.0006	0.0004	0.0006	0.0005	0.0009	0.0007	0.0010
2024 Q4	0.0005	0.0007	0.0004	0.0009	0.0004	0.0011	0.0006	0.0012
weekday:								
Monday	0	0	0	0	0.0000	0.0001	0.0000	0.0001
Tuesday	0.0000	0.0001	0	0	0.0000	0.0001	0.0000	0.0002
Wednesday	0.0000	0.0004	0	0	0.0000	0.0002	0.0000	0.0001
Thursday	0.0000	0.0001	0	0	0.0000	0.0002	0.0001	0.0002
Friday	0.0000	0.0002	0.0000	0.0002	0.0000	0.0002	0.0000	0.0002
Saturday	0.0002	0.0006	0.0002	0.0008	0.0002	0.0015	0.0002	0.0015
created_after_ invasion	0.0078	0.0077	0.0085	0.0148	0.0082	0.0307	0.0065	0.0286
created_after_ invasion_created_ after_pandemic	0.0019	0.0068	0.0010	0.0063	0.0002	0.0009	0.0004	0.0016

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feature	no embeddings		TF-IDF		Word2Vec		BERT	
	perm	gain	perm	gain	perm	gain	perm	gain
created_after_ invasion_ population_total_ log	0.0010	0.0133	0.0010	0.0066	0.0012	0.0038	0.0010	0.0039
created_after_ pandemic	0.0724	0.1299	0.0503	0.1158	0.0650	0.1113	0.0653	0.1055
created_after_ pandemic_ population_total_ log	0.0251	0.0406	0.0347	0.0508	0.0426	0.0427	0.0434	0.0433

Note: perm is permutation importance; gini is mean decrease in impurity.